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RPG-GPT

Leveling up game dialogue with creative NLG

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Tiivistelmä – Referat – Abstract

While natural language generation (NLG) and large-language models (LLM) seem to be transforming many industries, video games have yet to be affected. This study investigates the potential of using NLG systems to generate dialogue for non-playable characters (NPCs) in role-playing games (RPGs). For this, dialogue data is extracted from six popular RPGs and is then used to fine-tune Microsoft's GODEL to create an "RPG chatbot" (RPG-GPT). Motivated by computational creativity frameworks, a survey and an interactive experiment were conducted to evaluate the creativity and the effectiveness of RPG-GPT in generating relevant and engaging responses to player input.

Survey respondents rated dialogues on a 5-point agree-disagree Likert scale, with questions related to e.g. the relevance of the NPC answers. Results indicate that RPG-GPT can provide relevant responses with a mean difference of game relevance of 3.93 vs. 3.85 of RPG-GPT ($p=0.0364$). Also, the participants of the interactive experiment reported engagement when interacting with RPG-GPT.

Overall, the results suggest that creative NLG has the potential to enhance gaming experiences through task-oriented game dialogue (TOGD) systems. In this framework, creative TOGD systems could solve a common issue where pre-written NPCs are unable to provide the specific information sought by players. Additionally, the study discusses a concept of how players through their interaction with the NLG models can expand the lore of a game, which is a new consideration for game designers and developers when implementing such systems. Future work could explore ways to incorporate external knowledge and context to improve the performance of a TOGD system.

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Chapter 1

Introduction

1.1 Motivation

Natural Language Generation (NLG) and large-language model (LLM) systems have been rapidly gaining popularity and are transforming the ways in which users interact with various applications. One of the most recent famous examples of this is how OpenAI's ChatGPT has caused many academic institutions to publish guidelines for its use (YLE News, 2023). While NLG systems are designed to generate human-like language and they seem to be transforming many industries, video games have yet to be affected. In fact, game developers could potentially utilize NLG to produce dialogue for non-playable characters (NPCs) in video games. However, the effectiveness of such systems in generating coherent and creative video game dialogue is still relatively unexplored.

It could be argued that video games are unaffected by the transformative effect of LLMs due to the lack of game design frameworks that can incorporate open ended text generation. Moreover, NPCs based on LLMs "are liable to go off on tangents, discussing topics outside of the game's world." (Bedingfield, 2023) Nevertheless, it is likely only a matter of time until we see some video games, especially role-playing games (RPGs), incorporating these types of systems. Motivated by this, this paper aims to explore how such systems could affect player experience, especially compared to traditional pre-written dialogues and narratives.

Developing an NLG system for RPG dialogue presents several considerations. In particular, the system must be capable of providing relevant and engaging responses to player input, potentially while maintaining consistency with the game's narrative and world-building. A successful system should enable players to feel immersed in the game world, and foster a connection with the story and characters encountered. It is, therefore, imperative to evaluate whether NLG systems can achieve a sense of player engagement through their generated dialogue. This can be measured through various indicators such as player concentration, immersion, and temporal dissociation. By assessing the ability of NLG systems to create an immersive player experience,

developers can determine the feasibility of utilizing such systems to generate dialogue in video games, specifically in the context of RPGs.

A comparison between an NLG system and traditional pre-written dialogue in games can help provide insights into the effectiveness of NLG systems. Although traditional pre-written dialogue has been widely used in game development, it can be limiting in terms of the number of responses and the level of interactivity it offers compared to an NLG system which could potentially increase player agency (Pöyhönen et al., 2022). Therefore, it is important to explore whether NLG systems can provide a more dynamic and personalized experience for players while considering whether there are any notable differences in player experience between the two approaches.

Computational creativity is a rapidly growing field that in this context has the potential to transform the way we think about creativity and its role in video games. While creativity is typically associated with people, there has been much development within the field of computational creativity about the potential for machines to exhibit creative behaviors. By developing computer systems that can generate novel and valuable artefacts, researchers in this field are paving the way for new forms of human-machine interaction. This raises questions about the extent to which NLG systems can create novel and interesting dialogues and story lines in video games. For this, the field of computational creativity can provide frameworks for evaluating the systems that generate such content, including models for novelty. By utilizing computational creativity in this way, one is able to evaluate and then improve the effectiveness of these types of NLG systems in generating engaging dialogue that enhances player experience. Overall, exploring the potential for NLG in video game dialogue provides a fascinating intersection between natural language processing, creativity, and game design, and it is a promising space for further investigation.

1.2 Research questions

Based on the motivation above related to NLG, computational creativity and video games, this paper presents the following research questions to examine how effective an NLG system is in generating coherent, relevant and engaging video game dialogue:

> 1) How well does the system perform in terms of generating responses that are relevant to the player's input?
>
> 2) How well does the system perform in terms of creating a feeling of engagement in the player?
>
> 3) To what extent does the system exhibit creativity?
>
> 4) What kind of differences in player experience is there between an NLG system and traditional pre-written dialogue in games?

These questions have been selected in order to investigate player experience when interacting with generative AI systems, examining how these systems compare to traditional pre-written multiple choice game dialogues, and explore the feasibility of using these types of generative models in video games in the future. To answer these questions, this study will extract a video game dialogue data set and use it to fine-tune a large language model (LLM.) This resulting model (RPG-GPT) will be evaluated based on computational creativity frameworks, using two human evaluation methods: a survey and a qualitative experiment, where participants interact with both the game system and the generative model.

Chapter 2

Background

2.1 Dialogue in video role-playing games

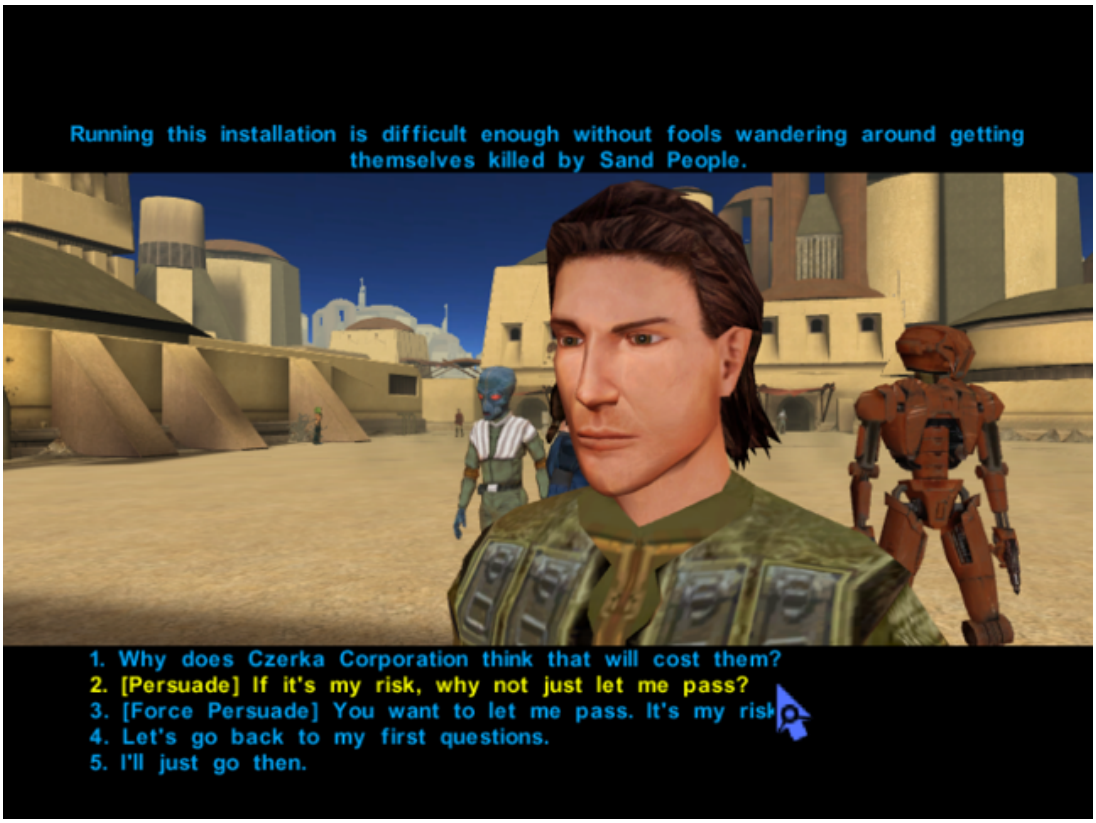


Figure 2.1: Image: Screenshot (Pöyhönen et al., 2022) of dialogue from Knights of the Old Republic 1 (BioWare, 2003).

Standard dialogue in role-playing games (RPGs) is pre-written in the sense that the player chooses among several pre-defined options on how to respond to a dialogue turn spoken by a non-playing character (NPC.) However, in some games, the dialogue system is more dynamic

and allows for more player agency. This is achieved via branching dialogue trees, where the new dialogue options presented to the player often depend on the previous one that the player has chosen. In this way, the dialogue can branch out in multiple ways. Moreover, the games using this type of dialogue system are often designed in such a way, that the player can have an effect on the story through dialogue and other actions in-game.

The player's choices and actions throughout the dialogues and game may help in creating a more immersive and interactive experience. Additionally, some games may also incorporate dialogue skill check mechanics such as persuasion or intimidation, which can further affect the outcome of the dialogue. By the gamification of dialogue options, the player may experience a greater sense of agency and immersion in the dialogue.

2.2 Dialogue generation

Dialogue systems (or chatbots) are often divided into two categories: task-oriented dialogue systems (TOD) and open-domain dialogue systems (OOD) (Ni et al., 2022, p. 2). Recently, these systems are often based on deep learning as they outperform rule-based ones. Whereas TOD systems are often used to solve specific problems for users, such as movie ticket or restaurant table bookings, the OOD systems aim to chat with users without the task and domain restrictions (Ritter et al., 2011). In both systems, it can generally be seen that the systems are a mapping of a user message to an agent response (Ni et al., 2022).

As noted by Ni et al. (2022) in their survey, "Humans associate current conversation context with their experiences and memories and produce meaningful related responses." In this way, external knowledge can be useful information when training and fine-tuning chatbots as further context for the dialogue. Moreover, while the OOD systems can often take into account the dialogue and dialogue turns that have already taken place as context, there is potential for a longer-term context with "long-term goals" as discovered by Xu et al. (2020). This approach "encourages the agent to converse with a long-term goal" Ni et al. (2022) In fact, this could be useful in "negotiation or story-telling chatbots" as a chatbot with a goal in guiding the player through e.g. a specific narrative could potentially be effective in generating video game dialogue.

2.3 Flow and immersion

Various measures and scales have been used to examine measure player experience during video game-play, though most are based on the work by Csikszentmihalyi et al. (2005) on flow theory. This theory is used to describe what an individual experiences during play, and thus, is a "central theoretical frame for understanding user experience in game-based settings." (Wiebe et al.,

2014, p. 124) Csikszentmihalyi's flow can be "thought of as a deep immersive experience that results from an individual engaging in a task." This is critical as the foundation for evaluating player experience during video game play, as it serves as the starting point for understanding further research that examines player behaviour and psychology.

It could be argued that the work by O'Brien and Toms (2010) in educational multimedia acted as a starting point for having Csikszentmihalyi's flow theory being adapted into video game engagement research, (Wiebe et al., 2014, p. 125) and their work was extended further by Wiebe et al. (2014) who focused their study "by investigating the use of their [O'Brien and Toms] User Engagement Scale (UES) in the context of game-based environments." (p. 1) With this study, the authors used exploratory factor analysis (EFA) to reveal four factors, out of which three aligned well with UES. This way, the authors have developed Flow theory closer towards conceptualizing player experience.

UES was measured as a 5-point Likert scale and comprised of 31 items (O'Brien and Toms, 2010) and was then modified by Wiebe et al. (2014) to better fit the items in the context of engagement in video games. For instance, the item "The time I spent shopping just slipped away" was modified into "The time I spent playing the game just slipped away." (p. 126). Out of the resulting items and the survey, the authors observed four factors to emerge. One of these factors was Focused Attention (FA), which had items (related to focused concentration, absorption and temporal dissociation) that captured self-reports of what the user was feeling while playing the game (e.g. "I was absorbed in my gaming task.") (p. 130.) Specifically, the FA items were distinct in terms of temporality from the other more reflective items, and also distinct from items related to visual assessments. In this sense, the results of this study illustrate FA as the factor that distinguishes items that represent engagement in a player, hence giving evidence that FA is a dimension of immersion in video games.

While the list of different dimensions of flow within the original flow theory was compiled from commonly reported sensations (Michailidis et al., 2018, p. 2), O'Brien and Toms "defined a number of both pragmatic and hedonic dimensions that influence user engagement." These dimensions include Aesthetics, Endurability, Felt Involvement, Focused Attention and Novelty. In this sense, it is important that also hedonic dimensions are included in the dimensions of UES, as Wiebe et al. (2014) argue that "while play can be task-oriented, these tasks can just as (or more) easily be driven by hedonic rather than pragmatic qualities of experience. Though play in both physical and virtual spaces is typically thought of as an activity or behaviour, what is most relevant to engagement is perhaps the user's perceived experience while at play and their response to this experience." (p. 125) While the original Flow theory contained pragmatic dimensions such as a sense of personal control or agency, merging of action and awareness, intense and focused concentration, it is critical to understand that hedonic qualities, such as aesthetics, can be significant contributors to engagement and flow, especially in the context of

video games and player engagement.

2.3.1 Player engagement

As Michailidis et al. (2018) argue, current research is inconclusive in that "the minimum requirements for an experience to qualify as flow remains unsettled (i.e. the necessary vs. the sufficient conditions; Swann et al. (2012).)" (p. 2) This could be seen as an issue in UES and the derived scale for video game contexts (UESz) by Wiebe et al. (2014), as it is unclear which factors or dimensions would constitute flow. In this sense, while the paper by Wiebe et al. (2014) does give evidence that Focused Attention seems to be an important dimension of flow, it is unclear whether this dimension is solely sufficient as a condition for flow. Possibly, in order to achieve flow within an experience of playing a video game, a player would likely also have to experience other UESz factors observed by Wiebe et al. (2014).

Yet, in comparison between flow and immersion, Michailidis et al. (2018) give the following model: "... the model of an average immersive experience in video game playing can be reduced to the engagement and engrossment levels, whose characteristics are not considerably divergent from flow." (p. 2) With this model and with the authors' conclusion that "immersion and flow do not appear as conceptually distinct," (p. 5) it may seem contradictory that flow and immersion are equivalent and that either or both can be reduced to engagement levels. In this case, it could then be possible to claim that solely measuring a significant amount of engagement, or more specifically Focused Attention (FA), would be a sufficient condition for flow or immersion in a player. Nevertheless, as was observed before, it is merely unclear what the sufficient conditions are, and that it could in fact be possible to achieve player immersion solely via engaging the player.

2.4 Computational Creativity

Computational creativity is a rapidly growing field that focuses on the development of computer systems capable of generating creative artefacts. These artefacts can be anything from e.g. music to literature. As an example of computationally creative NLG, Hämäläinen and Alnajjar (2019c) present a creative poem generator for Finnish. Additionally, while computational creativity, and creativity in general, is often associated with more artistic contexts, there also exists work where it is applied in practical contexts, with e.g. generated factual news headlines (Koppatz et al., 2022).

There is a significant amount of the work in this field that is concerned with "defining creativity from a computational point of view." (Alnajjar et al., 2021) In this sense, the defining frameworks "take into account having something non-human, such as a computer, as a creative

agent than the theories explaining human creativity." (p. 5) With this, while computational creativity definitions and frameworks can take inspiration from human creativity, Colton (2009) argues that the two types of creativity approach creativity from different contexts, yet ultimately they look at the same phenomenon (Alnajjar et al., 2021, p. 3). Moreover, as Alnajjar et al. summarize, creativity can be different between cultures and individuals (Shao et al., 2019) and that generally, scholars "agree that creativity results in product that is novel and has value (Mumford, 2003)."

2.4.1 The creative tripod

In order to define computational creativity and what it would mean for a computational system to be creative, Colton (2012) provides a "creative tripod" framework that takes into account both the creative process as well as the creative output. For this, Colton discusses idea that software can be creative by identifying three main components: skill, appreciation and imagination. Colton posits that these three elements (or legs) are integral to the creative tripod model and are critical to understanding how computational systems can exhibit creativity. For this, skill refers to the ability to create the ideas, imagination refers to the ability to generate specifically novel ideas, and appreciation refers to the ability to evaluate the creative output and its process. Colton has another way to summarize the three legs: "Without skill, they would never produce anything; without appreciation, they would never produce anything of value; and without imagination, at best they would only produce pastiches of other people's work." (p. 5)

One reason for the development this framework is rooted in the notion that the research community was mainly focused on assessing creativity through the evaluation of the output or artefacts produced by a computational system (p 2.). Colton suggests that while this type of approach is more than adequate for certain types of creative systems, it can be limiting for others, as it ignores the processes involved in producing the creative output. This idea is further supported by the finding that consumers of art do not strictly separate the process and the artifact. (p. 1) Rather, the perception of how a piece of art was produced can influence their overall enjoyment of the artwork. With this, it is not only important that there exists the ability to create an artefact (skill) or that the artefact is novel (imagination), but the tripod also assumes a third leg: appreciation, in that a consumer of art often also considers the process of producing the artefact.

A concept that is important especially in the context of computational creativity and not simply creativity in general, is the fact that there can be multiple different parties - programmer, program and perceiver - that can contribute to the experience by contributing taking the role of one of the legs. (p. 5.) With this, software could be creative in that a user could appreciate the artefact, though in this case the software would not be autonomously creative.

Skill

In Colton's creative tripod model, skill is one of the three fundamental components of creativity. Skill refers to the artist or software having the basic ability to produce the work or artefact, and to apply the necessary techniques and procedures to create something new. For instance, software "simulating paint strokes to render the image" (p. 6) could be seen as skillful. Colton argues that skill is a necessary but not a sufficient condition for creativity; it provides the means to produce creative artefacts but does not guarantee their value or novelty. The development of skill could be seen as a process that involves the acquisition of new techniques and the refinement of existing ones.

Appreciation

Appreciation is defined as the ability to evaluate or judge the quality and value of creative artefacts and their underlying processes. In other words, it is the capacity to discern and recognize the worth and significance of a creative output. Therefore, appreciation is a crucial component of creativity, as it helps to ensure that creative ideas are recognized and valued.

Imagination

Imagination refers to the ability to generate novel and original ideas, concepts, and possibilities that have not been encountered before. For a more specific and robust framework, this study uses Boden's framework on creativity (2007), and is in fact also referred to by Colton (2012) as part of the creative tripod.

2.4.2 Boden's framework on creativity

Boden's framework on defining computational creativity (2007) outlines three types of creativity: combinatorial, exploratory, and transformational. Combinatorial creativity is the ability to recombine familiar ideas, resulting in novel combinations. Exploratory creativity involves exploring and discovering new ideas and possibilities within a given domain. Finally, transformational creativity involves introducing new concepts or ideas that fundamentally transform a domain, often leading to paradigm shifts. Boden also distinguishes between two types of creativity: P-creativity, which refers to psychological creativity, and H-creativity, or historical creativity. Additionally, Boden notes that creativity involves not only the generation of novel ideas but also the evaluation and selection of those ideas, as well as the implementation of those ideas in an appropriate way.

P- & H-creativity

Psychological creativity, also known as P-creativity, is the creativity that occurs at an individual level. It is the ability of a person to generate ideas or solutions that are novel to them. In this sense, what is novel to one person may not necessarily be novel to others. P-creativity can occur in different domains, including art, science, and technology. Boden emphasizes that P-creativity is not simply about coming up with random ideas, but it involves a process of exploration and transformation of existing ideas in one's own personal style. While P-creativity is often seen as less valued than historical creativity (H-creativity), it is an essential aspect of creative expression and can still be considered very creative. For example, Boden discusses an example of a twelve-year old child who writes Macbeth without reading the work before. (p. 2) This would still be considered very creative, even though Macbeth has already been written and is thus not H-creative.

On the other hand, H-creativity is characterized as producing something that is not only novel to the individual but also to the culture or society. It involves creating ideas, products, or solutions that have never existed before and are considered to be groundbreaking or revolutionary. H-creativity is closely tied to historical and cultural factors, as the definition of what is considered new and valuable changes over time and across different societies. For example, the inventions of the telephone and electricity could be considered groundbreaking examples of H-creativity, as they have transformed society and humankind. The development of the internet is another example of H-creativity, as it has revolutionized the way people access information and interact with each other on a global scale. H-creative ideas and products can have a significant impact on society and can shape the course of human history.

2.4.3 Computational creativity evaluation

In terms of evaluation, it is important to note that not all NLP and NLG tasks have gold labels for quantitative metrics, and thus may sometimes need to be evaluated qualitatively due to their subjective nature. This is particularly true in creative natural language generation (NLG) tasks such as generating poetry, jokes, or in the case of this study, writing dialogue for video games. In these cases, evaluating the success of a model is often subjective and relies on the opinions and preferences of human judges (Hämäläinen and Alnajjar, 2021a). Because of this, questionnaires and other human evaluation methods are often employed when evaluating creative NLP systems. Nevertheless, in order to produce results that can be validated and reproduced, researchers have to ensure that what they are evaluating is in line with what is modeled. Unfortunately, there are multiple papers and studies in which this does not occur, and Hämäläinen and Alnajjar (2021b) call this as the Great Misalignment Problem. In fact, the original paper observes only one paper out of 10 randomly sampled papers (published in ACL 2020) to have its problem definition,

method and evaluation to be aligned.

According to another survey by Hämäläinen, M. and Alnajjar, K. (2021a), the fields of computational creativity and NLP have "a lot to learn from other fields from other when conducting human evaluation." (p. 91). The authors reviewed in total 23 published papers (ICCC and INLG). Most of these papers had issues in that they e.g. "did not justify why they had evaluated certain parameters." Also, Hämäläinen and Alnajjar argue that "Quite often with creative text generation, human judgement is the only viable metric to measure the performance of a system." With this, it is important to properly define and motivate evaluation parameters for effectively assessing the performance of creative NLP systems.

While the most critical issues in the methodology of computational creativity evaluation concerns justifying and motivating one's research and evaluation questions - and aligning them with the methods and models - there are also other very significant oversights. For instance, "most of the papers were not clear about the actual evaluation questions used." (p. 89) This is rather alarming, especially considering "even small changes in survey questions can lead to different survey results." (Kalton and Schuman, 1982; de Bruin et al., 2011; Bruine de Bruin et al., 2012) In this sense, not revealing the survey questions is also problematic, since it is a standard practice to benchmark survey questions and look for questions that have already proven to be working well.

In conclusion, the purpose of discussing the examples and criticism of creative NLP and NLG evaluation is to demonstrate the awareness of the oversights that can occur in a paper such as this. Hopefully, this review of the literature provides a perspective through which one can examine the evaluation methodology used in this paper.

2.4.4 Creative language generation and video games

There has been some computational creativity research in applying conversational AI in the context of video games (Fraser et al., 2018; Stegeren and Myśliwiec, 2021), though most of these works have not been actualized in any published video games, and have had their focus in evaluating player experience, e.g. engagement or model creativity. For instance, Fraser et al. (2018) used a pipeline of speech recognition, a rule-based system implemented in AIML (Wallace, 2001) and sentiment analysis to investigate player engagement and whether it is affected by augmenting the interface "with emotional processing" (Fraser et al., 2018, p. 184). While the authors claimed and provided some evidence that the system that had more empathy and emotion was more engaging compared to the "emotions disabled" system of the same pipeline, (p. 183) there are some aspects that future research could address. Firstly, the conversational system the authors had created was specifically for a single NPC (the barkeeper Hulda) and the 16 participants were tasked to obtain three specific pieces of information from the NPC. In this sense, while the participants "were instructed to converse freely with the NPC," the setting for

the dialogue could be seen as somewhat restrictive.

The authors discussed the following in the introduction of the paper to motivate their research.

> "While such a system [pre-written & branching dialogue tree] is very simple and easy for the players to use and progress with, as very little thought must be put in on the player's part, it also appears quite unrealistic and can break a game's immersion in otherwise engaging game worlds." (p. 1)

This is problematic as there has been no verified source that player immersion can be disrupted due to the nature of pre-written dialogue, and in fact, the opposite could possibly be argued due to how critically well received some of these games are. Nevertheless, the more critical oversight in the paper is that the authors do not measure player engagement between this type of dialogue system and their conversational AI system. Instead, they compare two of their AI systems (emotion enabled and disabled). In this sense, they do not actually provide evidence whether conversational AI can improve player engagement over a traditional pre-written dialogue system, but only give relative results between their two systems. Moreover, the evaluation methodology applied in this paper has some issues. For instance, while the authors are measuring player enjoyment and immersion, there is no given definition or motivation for immersion. Additionally, immersion is discussed interchangeably with player engagement. While the paper can be commended for publishing the survey questions, it unfortunately seems that the paper suffers from the same misalignment problem (Hämäläinen and Alnajjar, 2021b) as discussed above.

A more recent paper doing creative NLG within the context of video games is "Fine-tuning GPT-2 on annotated RPG quests for NPC dialogue generation" by Stegeren and Myśliwiec (2021) where the authors "investigate the usability of GPT-2 for text generation for video games." (p. 1) Although the paper seems to have well defined methodology and approach, the one aspect that could be critiqued about the paper is that the model by the authors is not attempting to generate dialogue at all. In fact, the fine-tuned GPT-2 model takes pieces of information such as the quest objective, location and reward as input and generates a descriptive text based on the linguistic structure of the World of Warcraft quests (which are used as training data.) While this paper does a significant contribution in bringing a generative transformer model to the context of video games, the model does not necessarily increase e.g. player engagement in dialogue but does present a more dynamic way to generate specific kind of quest descriptions.

The work in the paper mentioned above has been continued by Värtinen et al. (2022). The authors improve on the previous paper as the paper by Stegeren and Myśliwiec (2021) "only involved 20 quest descriptions rated by 32 participants" (Värtinen et al., 2022, p. 2) whereas

the experiment by Värtinen, et al. contains "349 participants rating a total of 500 quest descriptions." While this is certainly an improvement in terms of evaluating the creativity of a model, there is still potentially the issue of misalignment, as even the authors still admit that "assessing creativity however is not easy" (p. 7.) and that "the novelty of generated artifacts however is not straight-forward, as perceived novelty is highly contingent on individual experience (McGregor, 2007)." In fact, the authors in their evaluation ask the respondents' agreement with a single statement on a 4-point Likert scale: "I would be happy to see this quest description in a video game." (p. 9) In this sense, the authors "focus on assessing the quality of the generated quests" (p. 7) and while they "complement ratings with open-ended questions to gather further information on what influenced our participants' assessment," it seems that the evaluation does not lead to deeper error analysis about the creative aspects of the model, and instead the authors make a claim that the model "marks a step forward in achieving less repetitive and formulaic computer-generated quests." (p. 10) While it might be so that defining more specific aspects (than subjective experience of novelty) about the model's creativity was out of scope for this paper, this type of conclusion, while commendable, might suffer from an issue described in a survey paper by Hämäläinen and Alnajjar (2021a). In this survey, the authors notice that many creative NLG evaluation methods perform evaluation only for the purpose of determining the quality of the model. This leaves deeper error analysis lacking, especially in terms of "concrete future directions in improving the system based on the evaluation results." (p. 85) While Värtinen et al. (2022) do find multiple future directions to improve their work, none of them seem to be based on the human evaluation results.

There is a paper by Hämäläinen and Alnajjar (2019d) in which the authors present a mod (a third-party game modification) to integrate a creative dialogue NPC into Fallout 4 (Bethesda Game Studios, 2015). In this way, they are able to functionally apply creative dialogue generation in a video game. While the mod adapts the creative dialogue in a contextual way, by for example, taking into account the state of the player (such as health points or charisma) and number of quests completed or monsters killed, the authors still point out the future work possibility to integrate a mod such as this with multiple or all characters in the game (p. 7.) Moreover, as Hämäläinen and Alnajjar discuss potential ways to model context more thoroughly, perhaps a solution is now offered through the most recent deep learning based dialogue systems, and specifically as external knowledge used in training in addition to the mapping of context and response. Furthermore, it would be interesting to actually measure and investigate whether player engagement could be improved through a mod like this.

The purpose of this analysis is to show that there exists some exciting work at the intersection of NLG and video games. In this sense, the criticism in this section is only to hopefully situate this paper more accurately in terms of the field, research questions and the evaluation methodology.

Chapter 3

Data

3.1 Video games as source data

The method of extracting data from video games is similar to previous work by Pöyhönen et al. (2022). In their work, the authors demonstrate that video games may contain semi-annotated data by the developers, and the viability of using this data for NLP tasks. For instance, the in-game dialogue may contain skill checks *(e.g. [Persuade] Come on, what harm is there telling me?)*, which through the game dialogue scripting could provide annotation. In other words, the scripted nature of in-game dialogues can be a valuable asset for NLP tasks such as sentiment analysis, named entity recognition, and persuasion detection, especially considering that creating data sets can often be slow or costly.

In addition to using the same data collected by Pöyhönen et al. (2022), the same methodology is used to extract more data from other games. Moreover, this paper also goes further to construct conversational data for the purpose of fine-tuning a generative pre-trained transformer (GPT) model.

Knights of the Old Republic 1 & 2

Knights of the Old Republic (KotOR) 1 & 2 (BioWare, 2003, 2012) are set in the Star Wars universe, taking place thousands of years before the events of the original Star Wars trilogy. The games were developed using the Infinity Engine, with KotOR 2 being developed by a different studio than KotOR 1. The story centers around a conflict between the Jedi Order and the Sith, with the player character taking on the role of a Jedi Knight or Sith Lord, depending on their choices. The games explore themes such as redemption, betrayal, and the nature of the Force. Both games are widely regarded as some of the best Star Wars RPG games ever made and have had a lasting impact on the franchise.

Neverwinter Nights

Neverwinter Nights is a role-playing video game that utilizes the same game engine as Knights of the Old Republic 1 & 2. However, unlike the latter, Neverwinter Nights is set in the fantasy universe of Forgotten Realms. This universe is widely recognized as a standard setting for the tabletop role-playing game Dungeons and Dragons (D&D.) The world of Neverwinter Nights offers a diverse setting to explore with a multitude of races, cultures, and societies. The game is primarily situated in the city of Neverwinter, located in the Sword Coast region, an area renowned for its ports, swamps, and ancient ruins. The world is replete with a variety of creatures, both friendly and hostile.

Pillars of Eternity 1 & 2

Pillars of Eternity (PoE) games (2015; 2018) are also developed by Obsidian Entertainment, though they are set in their own setting. The player plays a Watcher character which can peer into other people's souls. The games are set in a fantasy world known as Eora. The world of Eora is rich in history, culture, and lore, with a variety of races and factions vying for power and control. Players can explore a vast and detailed world, filled with diverse environments and unique locations to discover. The game is known for its high level of player choice and consequence, allowing players to shape the world and story through their decisions.

Tyranny

Tyranny (Obsidian, 2016) is an RPG that shares the same game engine as the PoE games, but with its own unique universe. The game is set in a medieval-inspired fantasy world, where the player takes on the role of a Fatebinder, a high-ranking official in the army of Kyros the Overlord, a powerful and tyrannical ruler. The game's mechanics emphasize the player's ability to make choices that will affect the game's story and world. In particular, players can choose what kind of ruler they want to be and must enforce Kyros' laws to maintain order in the conquered territories. The game's setting is characterized by a dark and oppressive atmosphere, and it explores the consequences of tyranny and the struggle between good and evil.

3.2 Data processing

During the process of extracting dialogues from video games, it is to be noted that the format of the dialogue files varies depending on the game engine employed by the respective game. For instance, games that are built on the Infinity Engine, such as KotOR and NWN, utilize dialogue (.dlg) and talk (.tlk) files, which are subsequently transformed into XML format. Conversely,

games based on the Eternity Engine, including PoE and Tyranny, use dialogue files saved in the JSON format.

Infinity & Eternity engines

The method of extracting the text data from the Infinity Engine games is identical to the previously mentioned method by Pöyhönen et al. (2022), and in fact, uses the same data. In summary, tlk2xml (xoreos Team, 2022) is used to extract dialogue lines and their IDs from dialog.tlk files. In addition to the text, certain kinds of metadata such as the possible replies to a dialogue file is extracted from .dlg files, which in turn are also converted into XML structure. Each dialogue line is an entry with a list of possible reply IDs.

The method for the Eternity Engine games is similar, yet more straightforward, as no file conversions are required. The dialogue files are in JSON format, with files for both the textual data (dialogue lines) and the metadata (replies and IDs.)

Cleaning

A Python function is applied to perform various text cleaning operations on the dialogue lines. This function uses regular expressions to remove parentheses and other unnecessary characters that are not alphanumeric. Additionally, a further step in cleaning the data is to remove any unnecessary tags such as "StartAction", "StartCheck" or "Start" from the data. Additionally, games like Pillars of Eternity and Tyranny contain information about named entities which can also be removed as they are not needed for the analysis. In future work, this information could be used as external knowledge to a dialogue. To further clean the data, any rows with missing text are dropped (no dialogue.)

The games also contain instances where, in the case of KotOR, the player navigates a computer console (for example, to view and hack into camera feeds) or in the case of Eternity Engine games, make player character choices and set up the background story for their character. These types of "dialogues" have been filtered out both manually and using regex searches.

Special tokens

In the process of creating dialogue data for training a language model, it might be important to consider special tokens that may appear in the dialogue. One example of such special tokens are "skill checks." These are commonly used in video games to represent a character's ability in a certain aspect of the dialogue. These skill checks may appear in different forms or be named differently in different games, and thus it is useful to combine similar ones, for example, renaming "Intellect" to "Intelligence" or combining "Spellcraft" into "Magic" to ensure more instances of these in the data set and make them more interpretable for a language

model. Additionally, some names and player options have been converted into special tokens. For instance, a player may name their character at the beginning of these games, and NPCs in the PoE games might refer to the player character by their name. Because of this, PlayerName and the respective strings for the player's culture and race have all been converted into "<|playername|>", "<|playerrace|>" and "<|playerculture|>." Similarly, the PlayerShip string has been converted into a generic "the ship." With this conversion, the meaning in a dialogue turn does not change, as PlayerShip in the game is a name.

3.2.1 Creating the dialogue turn dataset

This dataset and format is very similar to the Multilingual Persuasion dataset by Pöyhönen et al. (2022). For the purpose of this paper, the important columns are Response ID, Speaker ID, a list of possible replies, and the string of the dialogue line.

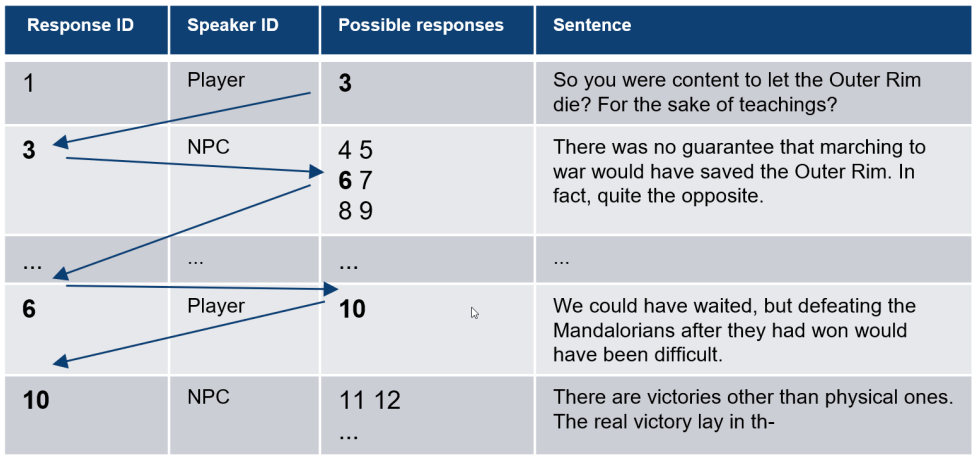


Figure 3.1: Illustrating the process of creating dialogues from list of responses

The process of creating the dialogue "trees" involves the use of a recursive function in Python. By considering the various possible ways in which a conversation can unfold, this method is able to construct a comprehensive representation of all the different ways that a dialogue can unfold within a conversation in a specific game scenario. The function is used to retrieve all replies to a given response ID, where each reply can have multiple layers of sub-replies. This function starts with a given response ID as the starting point, and then proceeds to check all possible replies to that response ID. Once a reply is selected, the process is repeated by checking all possible replies for the chosen reply's response ID, and so on. This continues until there are no more replies to the current conversation "path" or until a response ID has already been visited (to avoid infinite loops). The function then returns a list of all possible dialogues (that do not contain repeating dialogue turns) in the conversation tree. This process is illustrated in Figure 3.1.

3.2.2 Prompts and completions

Prompt-completion pairs are then constructed from each of the resulting dialogue paths. For this, each dialogue turn (starting from the last turn in a dialogue) is checked whether it is uttered by an NPC. If it is, the response is set as the completion, and the remaining dialogue as prompt. The process continues for the remaining dialogue, in that the next NPC dialogue turn is set as a new completion, and again the remaining dialogue is set as the prompt.

Prompt	Completion
Player: So you were content to let the Outer Rim die? For the sake of teachings? NPC: There was no guarantee that marching to war would have saved the Outer Rim. In fact, quite the opposite. Player: We could have waited, but defeating the Mandalorians after they had won would have been difficult.	NPC: There are victories other than physical ones. The real victory lay in th-
Player: So you were content to let the Outer Rim die? For the sake of teachings?	NPC: There was no guarantee that marching to war would have saved the Outer Rim. In fact, quite the opposite.

Figure 3.2: Resulting Example data from Figure 3.1

As an example, Figure 3.1 shows a dialogue that goes from Response ID of 1 to 3, and then to 6 and then finally to 10. Since the response ID 10 is an NPC dialogue line, so that line is set as the completion and the rest of the dialogue lines (IDs 1, 2, 3 and 6) as prompt. Continuing in the same dialogue, the response ID 3 is also uttered by an NPC. This is dialogue line is set as a completion and the remaining dialogue (response ID 1) as the prompt. The resulting two prompt-completion pairs can be seen in Figure 3.2. This type of data structure will be used for the purpose of fine-tuning the generative model.

The resulting data set consists of 847,038 prompt-completion (or context-response) pairs, which totals to approximately 128 million words. These pairs are constructed from approximately 350,000 dialogue turns. It contains a variety of different types of dialogues, from the player asking information from a variety NPCs, having narrative heavy dialogues with NPCs, etc. Many of these types of game dialogues used for fine-tuning the generative model can be seen in the Results section 5.1.3 Mean differences between game and RPG-GPT.

3.3 Ethical concerns

The use of data without explicit permission or consent raises ethical concerns, particularly when the data is creative work. In recent years, there have been several high-profile cases of AI models using copyrighted material without the artists' permission, such as the case of DALL-E (Ramesh et al., 2022) using artwork without the permission of the original artists. The same ethical concerns arise in the use of video game dialogue as training data, as this data is created by creative writers and game developers.

Using this data without permission from the original creators raises issues of intellectual property rights and fair use. Additionally, there may be concerns around the potential exploitation of the original creators' work. It is important to consider the impact on the original creators and whether their work is being used in a way that aligns with their values and intentions.

As a result of these ethical concerns, the data and models are not released publicly. While this may limit the availability of the research, it is a responsible decision that respects the rights of the original creators and helps to ensure that the work is used in a more ethical and responsible manner. Researchers working with similar data should consider the ethical implications of their work and take steps to ensure that they are using the data in an appropriate and responsible manner. In a specific case, it might be possible to have the data or the model available upon request and for research purposes only, although (and depending on any future licence) this might require a discussion with the holders of the creative rights.

Currently, there are discussions and legislation being worked on to address the legislative concerns around the use of artificial intelligence and machine learning models, particularly in relation to data ownership. This uncertain state of legislation and ongoing discussions are addressed by refraining from a free-access release of data and models together with this work.

Chapter 4

Methodology

4.1 GODEL

The text generation model used in this paper is GODEL Large V1.1 released by Microsoft in 2022. GODEL is a fully open-sourced pre-trained model for "goal-directed dialogues." GODEL has been trained on "551M multi-turn dialogs from Reddit discussion thread and 5M instruction and knowledge grounded dialogues." (Peng et al., 2022) In addition to Reddit discussion threads, GODEL has been trained with corpora that allow for grounded dialogue. These corpora are MS MARCO (Nguyen et al., 2016), DSTC7 Task 2 corpus (Galley et al., 2019), The Schema guided dialog (Chen et al., 2020) and UnifiedQA (Khashabi et al., 2020) (p. 3.)

GODEL Large is initiated from T5-Large and is based on the HuggingFace (Wolf et al., 2020) repository version with "BPE tokenization as implemented in HuggingFace" which is an algorithm used to compress data when pretraining GPT models. GODEL Large has 770M parameters and 1024 dimensional embeddings.

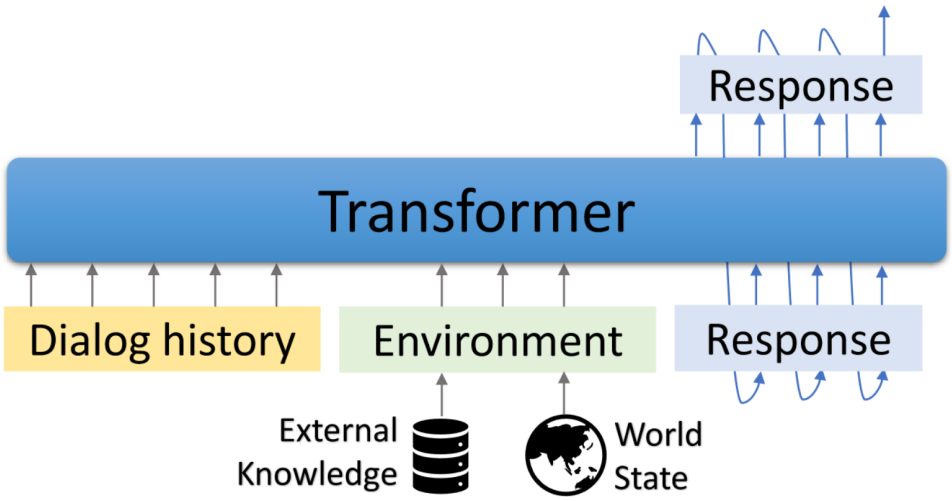


Figure 4.1: Overview of GODEL (Peng et al., 2022) architecture

GODEL has been trained for "response generation grounded in external text, which allows more effective fine-tuning on dialog tasks" and "can be efficiently fine-tuned and adapted to accomplish a new dialog task with a handful of task-specific dialogs." In this sense, this model should be suitable as it is then fine-tuned with video game dialogue data.

User : I would like to find an expensive restaurant that serves Chinese food. System : sure, which area do you prefer ? User : Bellevue downtown. <|environment|> Multiple expensive Chinese restaurants in Bellevue downtown. => There are multiple restaurants meet your requirement. peony kitchen is a great Chinese Restaurant. Would you like to book a table there?

Figure 4.2: A sample training instance, where conversation history is the text before the <|environment|> special tag, grounding is the text after the special tag, and the response comes after the "=>" symbol. (p. 3)

The sample training instance used in GODEL pre-training (seen in Figure 4.2) could be represented as in the following way:

Context - The context from session beginning to current turn.
Knowledge - External or environment state represented in plain text.
Response - The target agent response. It can be a template, an API call or natural language.

With this, Peng et al. (2022) describe the following algorithm in pre-training GODEL: "The open-domain goal-directed dialog generation task can be described as generating a target sentence T = (t1, · · · , tN) given a source dialog context S = (s1, · · · , sN) and an environment E, with the conditional probability of P(T|S, E) being the product of a series of conditional probabilities:"

$$p(T|S,E) = \prod_{n=1}^{N} p(t_n|t_1, \cdots, t_{n-1}, S, E)$$

GODEL is trained for 10 epochs and the best model is selected based on a validation set. (p. 4.)

4.1.1 Fine-tuning

The RPG-GPT was fine-tuned using specific hyper-parameters, including a total of 10 training epochs, a logging frequency of every 1000 steps, a training batch size of 128 per device, a weight

decay of 0.01, a logging directory, a saving strategy of "epoch", and a gradient accumulation step of 1. Additionally, gradient checkpointing was enabled and adamw_torch (Paszke et al., 2019) was chosen as the optimization algorithm.

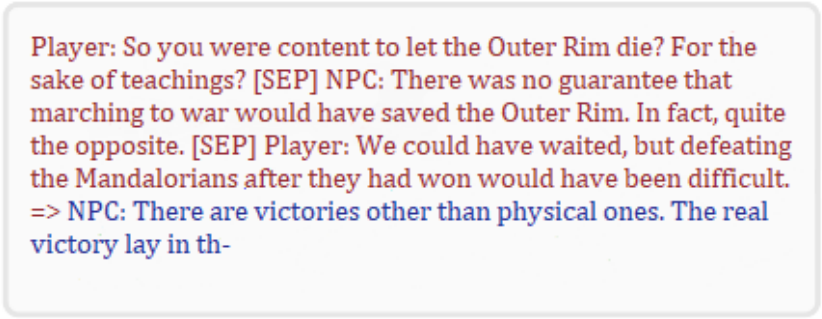


Figure 4.3: A sample fine-tuning example, where the prompt is followed by the => symbol which marks the following text as the completion.

Figure 4.3 shows an example fine-tuning instance. The turns are separated with the [SEP] token and the "Player:" and "NPC:" tokens are added as special tokens into the vocabulary. This is to further encourage the model to differentiate between the player character and NPC dialogue turns. A previous experiment with Microsoft's previous model, DialoGPT, showed some preliminary results in which the fine-tuned model would sometimes start producing responses that were exclusive to player characters, e.g. "I'd like to ask about something else." With this, and having only NPC dialogue turns in the completion (or response) the aim is to ensure that the resulting fine-tuned model would only act as the NPC.

After saving a model after each epoch, some qualitative testing was performed on each models. For this, the author had tens of dialogues with each epoch, and the best in the author's subjective opinion was selected. Based on these tests, the selected model had been trained for 4 epochs. Many of these dialogues can be found in the appendix.

The fine-tuning process used a custom DialogueDataset class to preprocess the input data. The dataset was designed to include a prompt (context) and a completion (response), while external knowledge for each instance was skipped. The tokenizer was set to truncate the prompt from the left side and the completion from the right side, with a maximum length of 256. This was done to ensure that the most recent context was included in the prompt, in that the last tokens uttered in the dialogue line were preserved and the first were left out. The encoded input and output pairs were returned as a dictionary containing the input ids and label ids, which were then used as input to the model during fine-tuning.

While GODEL has been pre-trained as a task-oriented dialogue system (goal directed dialogues), it could be seen that through fine-tuning, the system transforms into an open-domain dialogue system. This is because there is a relatively mixed set of instances and dialogues that

do not contain a specific "problem" that a fine-tuned model would be solving for the player.

During inference, the RPG-GPT model was set to temperature of 0.9, and a repetition penalty of 1,

4.2 Evaluation

In the case of video games, creativity plays a significant role in various aspects of game development, such as narrative and story. Therefore, evaluating the creativity of the model becomes an important aspect in determining its value. Motivated by this, this study aims to base the evaluation on existing frameworks of creativity which can in turn be used to assess the system's ability to produce creative dialogue effectively. These frameworks and the research questions are used to motivate both the evaluation questions and parameters.

4.2.1 Model creativity

While the creativity of the system is in the main focus as evaluation is performed, the practical functionality of this system can also be evaluated by examining the individual components that define the overall creativity of the system or whether the system is creative in the first place. For this, the creative tripod and its three legs are utilized: skill, appreciation and imagination. The first leg, skill, represents the technical abilities of the system, such as its ability to generate grammatically correct and relevant responses. The second leg, appreciation, represents the system's ability to engage and immerse the player. The third leg, imagination, represents the system's ability to generate novel responses and stories, which is a key aspect of creativity, as further defined by P- and H-creativity. In this way, evaluating the system in terms of these three legs might lead to a more comprehensive insight of both the model functionality and its ability to contribute to the creative process in a video game environment.

Skill

Within the context of this NLG system, skill is defined as the basic ability to converse in dialogue by having relevant answers to the player or user input. A key aspect of this leg is the system's ability to to generate responses that might create a feeling of inter-subjectivity in the plyer. Since it could be argued that an AI does not actually have an understanding of the player's subjectivity, it is important to emphasize that the player's feeling is closer to an "illusion." Overall, this is an important leg of the system, as it ensures that the player feels like they are having a conversation with the system, rather than simply receiving arbitrary responses from a generative AI model.

Skill in this context represents the system's ability to respond to the intent of the player's input in a manner that is relevant in two potential ways: immediate relevance and eventual relevance. In order to connect this more closely to what creative writers would consider as skillful in writing dialogue, the dialogue turns and responses should specifically be immediately relevant and should not seem arbitrary. Still, the responses can be surprising, but they still have to feel like they relate to the immediately previous player inputs. Nevertheless, if the responses are extremely immediately relevant, they could be seen as "boring" or meaningless (e.g. the word "Yes." as an answer to a long and complex question.) In this way, through relevance, the player would get a feeling that the system "understands" the player input and has a response according to what the player means to communicate. Without this, we can judge the system to be non-functional, as this is the basic level operation of the system and a necessary condition for creativity. Whether the player enjoys the output of the system and is feeling engaged in the dialogue is left to the appreciation leg.

There also exists relevance that is eventual, and here it would signify some kind of element in the dialogue response that has a meaning later on in the story or dialogue. An example of this is the concept of Chekhov's gun, where for example, if there is a gun in the narrative, there must be a reason for it, e.g. it being fired later. NLP and NLG methods have had trouble with this type of relevance (Tikhonov and Yamshchikov, 2021) and thus this kind of relevance is out of scope for this paper. There could be a potential way in including this type of relevance in external knowledge during a fine-tuning phase of GODEL, but it is not modeled in the current data in any way.

The research question, 1) "How well does the system perform in terms of generating responses that are relevant to the player's input?", is evaluated in such a way, that measuring and analysing the model's skill is directly equivalent to answering this research question.

Appreciation

In the context of the creative tripod framework, video games and this study, appreciation is modeled as player engagement. In this way, appreciation refers to the player's subjective experience of being engaged and absorbed in the dialogue. To model this type of engagement, this thesis proposes the use of Focused Attention (FA) (O'Brien and Toms, 2010), and specifically, the UESz version that has been developed further into the context of video games (Wiebe et al., 2014). By modeling player engagement, it is possible to evaluate and measure of this critical aspect of player experience. With this methodology, since the role for appreciation is assigned to the player, it's to be noted that the generative model will not be autonomously creative.

While Colton argues that for appreciation, the technique used to create a work of art might affect the consumer's opinion of it, in this paper this idea is somewhat inversed: if the player is engaged in the story and dialogue, and in this sense is not thinking about whether the dialogue

system is pre-written or generated, it could be argued that the player is then more likely to experience FA. In this sense, they might not want to be consciously aware of the method that there is a feature in the game where the AI is generating dialogue responses. If in fact players are thinking during the gameplay whether the dialogue responses are pre-written or generated by AI, they would be feeling less engaged or their immersion could be distracted, and hence appreciate the dialogue less. Nevertheless, if a player outside of the gameplay learns that the dialogue was generated by the AI, it could be seen as the type of appreciation where a consumer of art is evaluating the method and technique in addition to the work (video game or dialogue) itself after the fact.

Similar to the relationship between skill and research question 1), appreciation is evaluated and discussed as research question 2) "How well does the system perform in terms of creating a feeling of engagement in the player?"

Imagination & P-creativity

The third leg of the creative tripod, imagination, is concerned with the system's ability to generate new and unique responses that are not simply based on existing training data. This is important, as it ensures that the system is able to generate responses that are truly creative, and not just a repetition of pre-existing information. The ability to generate novel and unexpected responses is key to the system's ability to contribute creatively in a video game environment. In this sense, imagination can be evaluated by measuring the model's ability to produce dialogue or dialogue turns that cannot be found in the training data. This is a crucial aspect of the system's functionality, as it allows the system to generate unique and creative responses that can contribute to the narrative or story of the game.

It is worth noting that the use of H-creativity in the context of this paper would require a fundamental shift in the way that video game dialogue is currently produced and implemented. Therefore, the focus in this paper is on P-creativity, which is more directly applicable and in scope of this project. It is more difficult to conceptualize H-creativity, and it will not be discussed extensively here, as it is quite far from the scope of this paper. Nevertheless, some could argue that a kind of H-creativity could be seen as the ability to generate such dialogue, that hints towards completely new quests or narrative within the video game that the player is having dialogue in. With a more typical and traditional definition of H-creativity, the output dialogue could be seen as H-creative if the dialogue does not appear anywhere in any other video game and is in fact a completely new type of structure of dialogue or narrative. This definition is not useful in this context and is out of scope of this study.

While there is no direct research question regarding the system's ability to produce novel answers, the overall creativity of the model is assessed, as per the research question 3) "To what extent does the system exhibit creativity?" In this sense, all three legs of the tripod are necessary

conditions to consider the model creative.

4.2.2 Human evaluation

In order to evaluate the performance of the fine-tuned generative model in producing novel, relevant and engaging dialogue turns and responses, the creative and subjective nature of the task is taken into account. Traditional quantitative methods of natural language generation (NLG) evaluation, such as BLEU (Papineni, 2002), are less suitable in this case as there is no fixed target sentence or gold label to compare the model's output against. Nevertheless, BLEU has in fact been used in some computational creativity studies to assess how novel responses are compared to the training data (Hämäläinen and Alnajjar, 2019a,b). For this study, however, the model's skill and appreciation are in the evaluation focus, and hence the human evaluation methods are used instead. While using BLEU to measure novelty could be suitable in this study to measure the model's imagination, it is out of scope and left for future work. Instead, this study relies on human evaluation to motivate our evaluation methodology. For this, the study uses human evaluation in two ways: a survey and an interactive experiment. In this way, the survey produces more quantitative results focused on the fine-tuned model's skill, whereas the interactive user study aims to assess the level of engagement. Hopefully, these two evaluation methods are able to support each other in order to find more meaningful results.

Survey

The survey utilized seven dialogues extracted from the game and seven dialogues generated by an AI model. The game and AI dialogue pairs come from the same contexts which were randomly pooled in order to reduce author bias. Additionally, some of the dialogues created by the model were conversations between the generative model and the interactive study participants. The survey was published in Qualtrics.com (Qualtrics, 2023) which allows for shuffling the dialogues (one dialogue per section.) By shuffling the dialogues (not the questions), there is a reduction in priming - which is an effect when a stimulus affects the next experienced stimulus. (Henson, 2009) In this sense, a previous survey dialogue would affect a respondent's answers in the following one. With this, the order of dialogues will not affect the overall ratings of dialogues from the game system compared to the generated ones.

The survey was distributed in the subreddits r/SampleSize, r/SurveyExchange, and several Discord servers relevant to video games and AI. Also, the online research platform SurveyCircle was used for the recruitment of survey participants (SurveyCircle, 2023). Lastly, author's personal social media channels were also utilized to advertise the survey, such as Facebook, LinkedIn and Instagram.

All respondents were asked whether they had played any of the six games used in the source

data (KotOR 1 & 2, PoE 1 & 2, NWN, and Tyranny) or any other RPG video games. Then, using a five-point scale, the respondents were asked to evaluate their level of agreement or disagreement with statements related to relevance (relevance and intention), focused concentration (attention) and temporal dissociation (drawn). Specifically, they were asked to evaluate the following statements:

a) I saw that A had an understanding of B's intention. *(intention)*

b) I think A's responses were relevant to what B said. *(relevance)*

c) The dialogue captured my attention and held it throughout. *(attention)*

d) I was drawn into the dialogue. *(drawn)*

Each dialogue was followed by these four questions, with two questions evaluating relevance (a and b) - how relevant the answers of the model (both game and generative AI) are to the player input - and two measuring Focused Attention (FA) (c and d), with c) measuring focused concentration and d) measuring temporal dissociation. As observed by Wiebe et al. (2014) "Previously cited works on the development of a framework of user engagement note the importance of using multiple measures of engagement to capture behavioural and psychological aspects of engagement, but also to capture the differing facets of psychological engagement", this survey attempts to measure the model skill and FA with two questions each rather than a single one.

The survey was tested qualitatively before publishing. Three colleagues of the author tested the flow of the survey, and the responses were timed. The survey questions were benchmarked first from UESz by Wiebe et al. (2014) and then reiterated for clarity and in terms of fitting the context of this survey better, in the sense that the respondents will be reading dialogues and not actually playing a video game. The goal during reiteration was to make the survey questions as clear as possible while also keeping them concrete and ensuring that they were aligned with the research questions. Furthermore, the goal in making the survey questions and the dialogues public is to potentially make the results more transparent and reproducible.

A total of 86 survey responses were completed, with unfinished responses filtered out. Most respondents were unaware that some of the dialogues were generated by an AI, as believed they were taking part in a general survey focused on "Video game dialogue engagement & relevance." All (six) respondents who took part in the interactive experiment were aware that some of the dialogues were generated by an AI as well as the game system. Additionally, anyone in the close circle with the author most likely knew that the survey had something to do with AI. Nevertheless, the effect of "reading more into computer generated output than what the intention of the system was" (Veale, 2016) is hopefully more mitigated, as the majority of people are at least unaware of which dialogue is a generated one, and some are even unaware that any of

them are computer generated. Additionally, it is important to note that the respondents did not actually play a game, but instead were only presented with dialogue snippets, which may have impacted their engagement and overall perception of the dialogues. Moreover, any engagement or Focused Attention (FA) measured in the survey might be a different kind of FA compared to the interactive experiment, as the survey respondents are merely reading and not interacting, whereas the experiment participants are also interacting. Ideally, these measurements would be taken while the player is playing the game.

Interactive study

Six participants were involved in an interactive experiment in which they interacted with a game dialogue system and the fine-tuned RPG-GPT. The experiment was done remotely, with the participants connected through software that allows voice and text communication. Dialogue options and NPC responses were sent as text through the chat interface, with the participants also writing their inputs for the generative model through the same channel. The participants were aware which of the systems they were interacting with. The experiment aimed to compare the participants' experiences and feedback on the systems, with a focus on their level of engagement with the dialogues. During the experiment, the players were given different conversation scenarios to choose from when interacting with the generative model. As for the game dialogue system, the players were first given the context for the dialogue. Then, they were shown what the player character had initially said and the NPC's response to that. Finally, the participant was presented a choice between dialogue options that follow the NPC response.

The dialogue tree that the participants could navigate began with the following player input and NPC response:

> **Player:** Were you not tempted to help the innocent?
> **NPC:** Of course I was. But the Jedi teachings require we examine how we may best help them - action without reflection is not our way.

After reading the beginning of the dialogue, the participants were tasked to select one of the following player dialogue responses:

> **1.** So you were content to let the Outer Rim die? For the sake of teachings?
> **2.** Tell that to the millions who died in the war... lives we could have saved, if the Jedi had acted earlier.
> **3.** Our teachings do not mean we should stand by and watch others die.
> **4.** Hesitation and passivity are also strong emotions - ones we should guard against.
> **5.** I did not come here to debate the Council's decision.

Option 5 ends the dialogue, with the NPC responding "Very well. Your exile has given you some wisdom at least." whereas for the other options, the NPC would respond with "There was no guarantee that marching to war would have saved the Outer Rim." After this, there would again be options for the participant, and this would continue until the participant made a choice among the dialogue options a total of four times.

During the interactive experiment, each participant was asked open-ended questions related to their experience of both dialogue systems. The participants' feedback included their general experience of each system and the specific aspects they enjoyed or found lacking in each system. The open-ended questions allowed for more detailed feedback from the participants and provided insights into their personal experience with each system. Overall, this interactive experiment aimed to provide a comprehensive evaluation of the participants' experiences and feedback on both the game dialogue system and the generative model.

Previous studies have shown that qualitative surveys require 5-25 (Creswell, 1998) or at least six participants (Morse, 1994). The amount of participants in this interactive experiment is chosen accordingly as 6 in order to ensure the results are more meaningful. Only one of the participants had played (not much) a game from the list of games that consist of the source data. Still, the demographics are varied with age, gender and in terms of gaming experience. In fact, some of the respondents do not extensively play video games while among those that do play games, it varies how much they play RPGs specifically.

4.2.3 Analysis of survey results

Several methods are used to analyse the survey results. First, each of the seven contexts are taken and the Wilcoxon matched pairs signed rank test (Wilcoxon, 1945) is performed between the AI and game dialogues. This way, the means between the survey items (game vs. AI) can be compared while measuring whether they are statistically significant. It is to be noted that if an item is not statistically significant, it does not necessarily mean that the AI and game dialogue is "equal quality." In fact, it is only unclear that the samples come from different distributions. Second, Exploratory Factor Analysis (EFA) is performed to examine the underlying variables that can be used to generalize the survey items. With this, the approach will provide evidence whether specific survey questions are measuring the correct aspects of the model (skill and appreciation.)

Exploratory Factor Analysis

Exploratory factor analysis (EFA) (Fabrigar and Wegener, 2012) is a statistical technique used to identify the underlying structure of a set of variables. It is commonly employed when researchers are seeking to identify the underlying factors that explain the inter-correlations among

a set of observed variables. EFA can be used to test hypotheses about the number of underlying factors, and to identify which variables are most strongly associated with each factor. In this sense, the aim of EFA is to reduce a large set of variables into a smaller set of underlying factors that explain the covariation among the variables. The factors that emerge from an EFA are typically not directly observed or measured, but rather inferred from the relationships among the observed variables.

In order to provide evidence for the viability of measuring the model's skill (survey items a and b) as well as appreciation through Focused Attention (FA) (survey items c and d), exploratory factor analysis is performed. The aim of this analysis is to see whether the corresponding items group into their own factors. Then, it could be argued that the items are measuring a single "hidden" variable (skill or FA) that can then be evaluated in the context of their respective research question. In addition to the above, a correlation matrix is calculated first. This gives some initial insight into which survey items are mutually correlated.

Bartlett's test of sphericity (Bartlett, 1937) is used as a statistical measure used to determine whether a set of variables are appropriate for factor analysis. It tests the null hypothesis that the correlation matrix of the variables is an identity matrix, indicating that there are no correlations among the variables. A significant result on the Bartlett's test indicates that the variables are inter-correlated and that factor analysis is appropriate. The test calculates the determinant of the correlation matrix and compares it to the chi-square distribution with degrees of freedom calculated from the number of variables. If the determinant is significantly different from zero, then the variables are not independent and a factor analysis can be performed. Bartlett's test of sphericity is commonly used in exploratory factor analysis to evaluate the factorability of the data and ensure that the assumptions underlying the analysis are met.

Kaiser-Meyer-Olkin (KMO) (Kaiser, 1974) is also used as a measure of sampling adequacy and suitability of a dataset for factor analysis. Specifically, KMO measures the proportion of variance among the observed variables that is common variance, or variance that can be explained by underlying factors. KMO values range from 0 to 1, with values closer to 1 indicating greater sampling adequacy. A KMO value of 0.6 means that the items are suitable for factor analysis. KMO is useful because it takes into account the partial correlations between variables, which means that it is a more accurate measure of sampling adequacy than some other measures that only consider the correlations between variables.

Chapter 5

Results

The results of the survey and the interactive user experiment are presented here. For the survey, each dialogue context (game and AI dialogue pair) is presented with average means. Additionally, the comments from the experiment participants are summarized into their own section.

5.1 Survey

For the survey results, there are two supporting approaches to analyze them: exploratory factor analysis (EFA) and a correlation matrix. First, the correlation matrix is given to provide some initial insights into how the survey items are correlated. Then, a more in-depth factor analysis is performed to examine how the survey items relate to each other. Finally, once there is enough evidence for measurement of skill and Focused Attention (FA), the average means for each item and factor is given, with skill measuring the relevance of the answers provided by RPG-GPT and FA measuring the engagement and immersion levels of the reader.

45 survey respondents responded that they have not played any kind of RPGs. As approximately half of the respondents have not played any RPGs, there is some variety in terms of how creative dialogue writing is read and interpreted, as in this sense, is not biased towards only people who play a lot of video games.

5.1.1 Correlations

In the correlation matrix (Table 5.1) we can see the eight survey items that will be examined throughout this results chapter. AI and Game correspond to the method of which was used to produce the dialogue. Intention, relevance, attention and drawn correspond to the survey questions, (survey questions a, b c and d respectively.) With this, four aspects are examined: 1) whether intention and relevance are correlated, 2) whether attention and drawn are correlated, 3) whether AI and game items are correlated, and lastly 4) whether all relevance, intention,

	AI/rl	Game /rl	AI/in	Game/in	AI/at	Game/at	AI/dr	Game/dr
AI Relevance	1	0.09	**0.657**	0.084	0.472	0.076	0.468	0.032
Game Relevance	0.09	1	0.087	**0.703**	0.101	0.588	0.121	0.533
AI Intention	**0.657**	0.087	1	0.228	0.409	0.107	0.451	0.109
Game Intention	0.084	**0.703**	0.228	1	0.122	0.556	0.157	0.548
AI Attention	0.472	0.101	0.409	0.122	1	0.293	**0.813**	0.266
Game Attention	0.076	0.588	0.107	0.556	0.293	1	0.3	**0.815**
AI Drawn	0.468	0.121	0.451	0.157	0.813	0.3	1	0.326
Game Drawn	0.032	0.533	0.109	0.548	0.266	**0.815**	0.326	1

Table 5.1: Correlations between survey items. AI items are the survey items from AI dialogues, and Game items respectively from game dialogues. The most correlated item for each item is bolded.

attention and drawn are correlated.

AI relevance is the most correlated with AI intention, and the same is the case for the game items respectively. Similarly, AI attention is most correlated with AI drawn, and the same is again the case for the game items. As for 3) and 4), the items do not seem to be all inter-correlated. For instance, the AI items are more correlated between themselves rather than with their respective game items (e.g. AI relevance and Game relevance of 0.09.) The items between 1) and 2) are somewhat correlated within the same dialogue system. For instance, AI attention has a correlation of 0.472 with AI relevance. It is to be noted that the correlations between drawn and attention items (FA) is much higher than between the relevance and intention (skill) items.

Looking at the survey items in terms of their inter-correlations, it is evident that e.g. the AI relevance item (question b) "I think A's responses were relevant to what B said.") seems to be more correlated with other AI items rather than with game relevance (or game intention.) In this sense, an item is more correlated with whether it is from an AI or game dialogue, rather than the item being correlated with another item that measures the same expected dimension.

5.1.2 Factor Analysis

Calculating the Bartlett Sphericity reveals that the $p<0.001$ and chi is approximately 2187.89. Additionally, the KMO is 0.73, which suggests that the data is adequate for factor analysis.

According to Figure 5.1, it seems that there are two factors above the eigenvalue of 1, while a third factor is extremely close with a value of 0.97. Additionally, looking at the plot visually, it seems that the "elbow" would indicate that there could be four factors, and while this evidence is somewhat weaker as the fourth factor is at 0.5, this idea is still explored below.

First, when the survey items are divided only into two factors, we can see that there is a more clear distinction between game and GODEL items, instead of relevance and FA. This can be seen in Table 5.2.

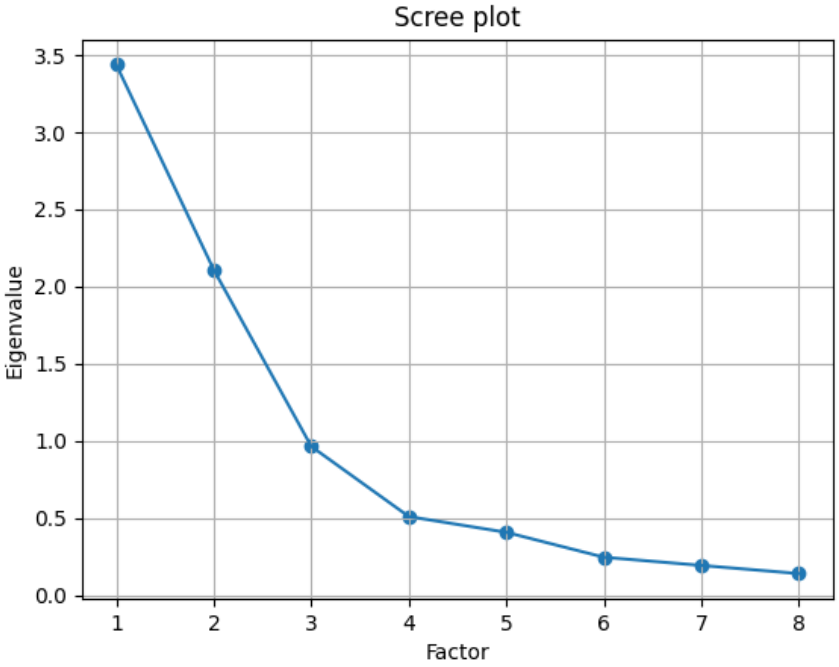


Figure 5.1: Screeplot of factors and eigenvalues. The factors in this graph are the four AI and the four game items (relevance, intention, attention and drawn)

	1	2
Relevance (AI)	0.013466	**0.689572**
Intention (AI)	0.063662	**0.636887**
Attention (AI)	0.152783	**0.791805**
Drawn (AI)	0.186793	**0.814876**
Relevance (Game)	**0.745068**	0.029286
Intention (Game)	**0.726879**	0.088094
Attention (Game)	**0.846200**	0.160158
Drawn (Game)	**0.812776**	0.151529

Table 5.2: Factor analysis with two factors

	1	2	3	4
Relevance (AI)	-0.053597	0.393375	0.071677	**0.584267**
Intention (AI)	0.066977	0.208789	0.054211	**0.974325**
Attention (AI)	0.297573	0.038682	**0.951893**	0.014993
Drawn (AI)	0.433506	0.008934	**0.598457**	0.154255
Relevance (Game)	0.133460	**0.905956**	0.028948	0.213461
Intention (Game)	0.199201	**0.800426**	0.027446	0.276833
Attention (Game)	**0.742265**	0.193507	0.381359	0.001621
Drawn (Game)	**0.918685**	0.158622	0.261960	-0.000114

Table 5.3: Factor analysis with four factors

When the items are divided into four factors, it is evident that they belong into factors that are combinations of dialogue method (AI/Game) and the measured aspect of the model or dialogue (skill/FA). In this sense, there is evidence that the survey was able to measure e.g. the skill of GODEL, via the two questions (a and b) in terms of the relevant answers that it produced. In this sense, according to the Table 5.3 it could be said that factor 1 represents FA on game dialogues, while factor 3 respectively for the game dialogues. Factor 2 seems to be game relevance or skill, while factor 4 is the measurement of the skill of the AI model.

5.1.3 Mean differences between game and RPG-GPT dialogues

The dialogues used in the survey as well as their corresponding means and standard deviations (game vs. AI) are listed in this subsection. First, for a general overview of the results, the mean differences are given across all contexts. Then, each following page will contain a context, the corresponding dialogues for that context and a table of the mean differences, calculated using the Wilcoxon paired signed rank test (Wilcoxon, 1945).

All contexts

	Statistic	p-value	Game	AI	Samples
All questions	549899.5	0.1196	3.7475 +- 1.1868	3.7006 +- 1.1274	2408
Relevance	32986.0	0.2343	3.9601 +- 1.1588	3.8904 +- 1.0721	602
Intention	29568.0	0.0754	3.9037 +- 1.1895	3.8007 +- 1.1334	602
Skill	124860.5	0.0364	3.9319 +- 1.1741	3.8455 +- 1.1036	1204
Attention	37959.0	0.9397	3.6096 +- 1.1518	3.5897 +- 1.136	602
Drawn	37647.0	0.9681	3.5166 +- 1.1895	3.5216 +- 1.1291	602
FA	151430.5	0.9889	3.5631 +- 1.1712	3.5556 +- 1.1326	1204

Table 5.4: Game vs AI (mean +- stddev) differences across all contexts and dialogues

Across all dialogues, the means are very similar. Also, the Skill factor, which consists of two items, is statistically significant. With this, it seems the AI is able to, in general, produce relevant answers. In fact, the mean is only slightly lower than the respective game mean (3.93 vs. 3.85.) For more specific analysis, a similar table to 5.4 will be computed for each dialogue context.

"Laethärn" context

Game dialogue

Player: Who's Laethärn? And what do you have against these farmers?
NPC: Dawnstars. Fleet Master Kō hired them to tend the fields that supply the mainland. But they started acting strange. Dreaming strange.
Player: You mean dreaming of things to come.
NPC: One of them, Laethärn, had a row with the fleet master over it. Left angry just before the colossus showed up.

AI dialogue

Player: Who's Laethärn? And what do you have against these farmers?
NPC: Dawnstars. They are a very dangerous breed, and they haven't been in the region in many, many years.
Player: What happened to them?
NPC: They were hunted by Kyros' forces, and many of their farms were destroyed.

	Statistic	p-value	Game	AI	Samples
All questions	9839.0	<0.001	3.1686 +- 1.21	3.4942 +- 1.13	344
Relevance	529.0	0.0246	3.1977 +- 1.2445	3.5581 +- 1.1743	86
Intention	502.5	0.005	3.1744 +- 1.229	3.593 +- 1.0779	86
Skill	2047.0	0.0004	3.186 +- 1.2332	3.5756 +- 1.1239	172
Attention	690.5	0.026	3.1628 +- 1.1768	3.4651 +- 1.155	86
Drawn	791.5	0.1242	3.1395 +- 1.2094	3.3605 +- 1.1158	86
FA	2929.5	0.0072	3.1512 +- 1.1898	3.4128 +- 1.1335	172

Table 5.5: Laethärn context means (Game vs AI mean +- standard deviation)

For the dialogues that begin with the dialogue turn of "Who's Laethärn? And what do you have against these farmers?", the AI version scored higher in all questions. All items except drawn are statistically significant. Arguably, the AI responses are somewhat more direct, whereas in the game version the NPC does not arguably answer the player question.

"How can I help?" context

Game dialogue

Player: How can I help?
NPC: Could I persuade you to watch out for Larro? I'd pay generously for your trouble.
Player: I take it your mother is someone important.
NPC: Ezzali Bardatto. The Vailian Trading Company leans on her to finance their mission. If only more of her offspring were as accomplished.

AI dialogue

Player: How can I help?
NPC: We are due to collect a special shipment for delivery into our port at Sayuki.
Player: I see, what is the shipment?
NPC: It is a cargo ship carrying crude iron from the mines at Ondra's Mortar.
Player: What is so special about crude iron?
NPC: It is special because it contains the essence of Harapo animancy. The metal has extraordinary magnetic properties, and when heated to high temperatures this can create incredible energy burst into living creatures!

	Statistic	p-value	Game	AI	Samples
All questions	6329.5	<0.001	3.1686 +- 1.2028	3.7035 +- 1.1628	344
Relevance	248.5	<0.001	3.2907 +- 1.2543	4.1512 +- 0.9764	86
Intention	313.0	0.0009	3.3605 +- 1.2453	3.9302 +- 1.1354	86
Skill	1109.5	<0.001	3.3256 +- 1.2467	4.0407 +- 1.0616	172
Attention	474.5	0.0111	3.0349 +- 1.1214	3.4186 +- 1.1424	86
Drawn	629.0	0.0462	2.9884 +- 1.1631	3.314 +- 1.1908	86
FA	2171.5	0.0013	3.0116 +- 1.1393	3.3663 +- 1.1646	172

Table 5.6: "Help" context means (Game vs AI mean +- standard deviation)

All items are statistically significant. The AI dialogue scores higher in all items. Both dialogues are related to a player receiving or asking for a task, or a "quest." As with the Laethärn context, the NPC answers with the generative model seem to be more direct.

"Menzzago" context

Game dialogue

Player: Who is this Menzzago?
NPC: One of the very first settlers, from what I've been told. So. What'll it be?
Player: Who's that in the helmet over there?
NPC: Eh? Ah, that'll be Copperhead.

AI dialogue

Player: Who is this Menzzago?
NPC: He's an ancient Huana scholar and an expert on the subject of the gods. He's been here for many decades, but his main interest is in the history of the island.
Player: What is your relationship with him?
NPC: When I first came here, I knew him as an erl. Then I became friends with him. Over time, we became close friends.
Player: Do you know where he disappeared?
NPC: No. It is a secret he keeps hidden. Only I know what it is.

	Statistic	p-value	Game	AI	Samples
All questions	8579.0	<0.001	3.4593 +- 1.1994	3.7616 +- 1.1124	344
Relevance	717.5	0.3767	3.8023 +- 1.1764	3.9302 +- 1.0824	86
Intention	518.0	0.1074	3.6163 +- 1.2478	3.8256 +- 1.16	86
Skill	2448.5	0.0875	3.7093 +- 1.2127	3.8779 +- 1.1198	172
Attention	475.0	0.0108	3.2326 +- 1.1548	3.6047 +- 1.1301	86
Drawn	448.5	0.0007	3.186 +- 1.1221	3.686 +- 1.0656	86
FA	1830.0	<0.001	3.2093 +- 1.1355	3.6453 +- 1.0959	172

Table 5.7: "Menzzago" context means (Game vs AI mean +- standard deviation)

For the "Menzzago" context, the only statistically significant results are for FA and its items (attention and drawn) with respondents preferring the AI dialogue across all items. The AI version is considerably longer with three NPC responses compared to the two turns in the game dialogue. Also, the turns themselves are somewhat longer in the AI dialogue. Arguably, the last NPC turn in the AI dialogue is somewhat odd ("secret he keeps hidden. Only I know what it is"), which might be the reason for a mixed result in terms of relevance.

"Refresh" context

Game dialogue

NPC: Why did you even return? To challenge the Council's decision on your exile? To defend your betrayal?
Player: Refresh my memory - why did the Council cast me out?
NPC: Why? Because you turned your back on us... and the Order.

AI dialogue

Player: Refresh my memory - why did you and the Jedi Council cast me out?
NPC: You followed Revan to war. And we knew about the terrible battle which would follow him if Malak returned and killed us both...
Player: So you were afraid? Is that it?
NPC: Afraid? Of what, the prospect of being withered away like my brother Revan... it was not an option!

	Statistic	p-value	Game	AI	Samples
All questions	10135.0	0.6096	3.8721 +- 1.1382	3.8198 +- 1.149	344
Relevance	486.0	0.2865	4.1744 +- 1.1082	4.0116 +- 1.0346	86
Intention	601.0	0.91	3.8837 +- 1.2021	3.8605 +- 1.1393	86
Skill	2141.0	0.3844	4.0291 +- 1.1619	3.936 +- 1.0877	172
Attention	789.0	0.9428	3.8023 +- 1.0496	3.7558 +- 1.2072	86
Drawn	675.0	0.8998	3.6279 +- 1.1381	3.6512 +- 1.1958	86
FA	2903.0	0.9008	3.7151 +- 1.095	3.7035 +- 1.1991	172

Table 5.8: "Refresh" context means (Game vs AI mean +- standard deviation)

For the "refresh" context there are no statistically significant differences between items. And the means are very close to each other. Nevertheless, this does not mean that the dialogues are equal in quality, only that we can not reject the null hypothesis (that the dialogues come from different samples.) The AI dialogue was generated with an interactive study participant.

"Merchant" context

Game dialogue

Player: Let me see what you've got for sale.
NPC: A: You want to see my inventory, or are you here to buy a utility droid? My prices are very reasonable, and all my droids come with a money back guarantee.
Player: Let me see your droids.
NPC: I've only got two droids left: a re-built T3-H8 unit for 50 credits and a top of the line T3-M4 model with all the options. But he's not for sale.
Player: I don't have 50 credits.
NPC: Well, I'm not going to drop the price on that model. At 50 credits it's already a bargain. If you can't afford it, you can't afford it.

AI dialogue

Player: Greetings, do you have some wares I could take a look at?
NPC: Yes, yes, of course! My inventory includes several lines of armor, weapons and armor accessories. You will find these to be very popular among our clients.
Player: Thank you. Do you know of any rumors in this town?
NPC: None that I know of. The only thing I know of is the black market that circulates around Defiance Bay. It is run by the same people who ran Defiance Bay during the Edict of Storms. I have not heard any other rumors about this place.
Player: What is the Edict of Storms?
NPC: The storms that destroyed our city were due in part to the coming of the Overlord. At the height of the storms, the Overlord placed an imposition on the power of the people of Defiance Bay. By the time the Overlord's armies had finished with their campaign, they had no choice but to surrender and leave the city in ruins.

	Statistic	p-value	Game	AI	Samples
All questions	6935.5	0.0012	4.0843 +- 1.0253	3.875 +- 1.0928	344
Relevance	370.5	0.1357	4.4535 +- 0.8211	4.2791 +- 0.8211	86
Intention	239.0	0.0057	4.5233 +- 0.8077	4.1977 +- 1.0154	86
Skill	1201.0	0.0028	4.4884 +- 0.8128	4.2384 +- 0.9216	172
Attention	609.5	0.1113	3.7674 +- 0.9783	3.5233 +- 1.1243	86
Drawn	587.5	0.4661	3.593 +- 1.1311	3.5 +- 1.1453	86
FA	2374.0	0.0969	3.6802 +- 1.058	3.5116 +- 1.1316	172

Table 5.9: "Merchant" context means (Game vs AI mean +- standard deviation)

The "Merchant" context, only the combined Skill factor is statistically significant. The means are relatively high for both of the dialogue methods.

"Teachings" context

Game dialogue

Player: Were you not tempted to help the innocent?
NPC: Of course I was. But the Jedi teachings require we examine how we may best help them - action without reflection is not our way.
Player: So you were content to let the Outer Rim die? For the sake of teachings?
NPC: There was no guarantee that marching to war would have saved the Outer Rim. In fact, quite the opposite.

AI dialogue

Player: So you were content to let the Outer Rim die? For the sake of teachings?
NPC: To die - to be destroyed - is the greatest lesson of all.
Player: I didn't take you for a dark jedi, Atris.
NPC: The greatest lesson of all is this - you are not the one who fell prey to Malak.

	Statistic	p-value	Game	AI	Samples
All questions	3420.0	<0.001	4.3401 +- 0.9119	3.5959 +- 1.0702	344
Relevance	264.0	<0.001	4.4302 +- 0.8195	3.4767 +- 1.0597	86
Intention	244.0	<0.001	4.3837 +- 0.8966	3.5233 +- 1.1655	86
Skill	1005.5	<0.001	4.407 +- 0.8567	3.5 +- 1.1108	172
Attention	186.5	<0.001	4.2791 +- 0.9536	3.7674 +- 1.0253	86
Drawn	179.5	<0.001	4.2674 +- 0.9753	3.6163 +- 1.0194	86
FA	726.0	<0.001	4.2733 +- 0.9617	3.6919 +- 1.0222	172

Table 5.10: "Teachings" context means (Game vs AI mean +- standard deviation)

For the "Teachings" context the respondents quite clearly prefer the game version of the dialogue. All the responses are statistically very significant, with all p-values being lower than 0.001, while the mean for AI is around 3.6 and for the game dialogue the mean is 4.3. Notably, there is some repetition in the answers of the generative model ("The greatest lesson of all..") which might be the reason for lower FA or Skill for the AI dialogue.

"Eothas" context

Game dialogue

Player: Let me ask you something else.
NPC: If you like.
Player: What are your thoughts on Eothas at the moment?
NPC: Doesn't seem much different than Thaos and Woedica, does he? Powerful beings thinking they know what's best for us. Not caring the costs we pay. Starting to think we'd be better off without the gods. At least we'd know the messes were our own.

AI dialogue

Player: What are your thoughts on Eothas at the moment?
NPC: I don't know. My thoughts are mixed, as you can tell from my recent travels to the Deadfire...
Player: After that your thoughts are mixed? How?
NPC: My emotions are tangled. I've been trying to get rid of them, but now they have so many layers on top that it feels like peeling back different skins...
Player: Are you okay? How did you think about Eothas before?
NPC: I didn't have a problem with him. He wasn't in control of his actions...
Player: Do you mean someone controlled him?
NPC: No, obviously not.

	Statistic	p-value	Game	AI	Samples
All questions	5599.0	<0.001	4.1395 +- 0.9828	3.6541 +- 1.135	344
Relevance	283.0	0.0005	4.3721 +- 0.8124	3.8256 +- 1.1082	86
Intention	207.5	<0.001	4.3837 +- 0.8425	3.6744 +- 1.1318	86
Skill	962.0	<0.001	4.3779 +- 0.8252	3.75 +- 1.1193	172
Attention	520.5	0.0073	3.9884 +- 1.0459	3.593 +- 1.1518	86
Drawn	445.5	0.0358	3.814 +- 1.0902	3.5233 +- 1.1451	86
FA	1910.5	0.0007	3.9012 +- 1.0687	3.5581 +- 1.1456	172

Table 5.11: "Eothas" context means (Game vs AI mean +- standard deviation)

Similarly to the "Teachings" dialogues, the game version scores much higher (avg 4.1 vs 3.6) with all items being statistically significant. The AI dialogue is generated by an interactive study participant.

5.2 Interactive study

This section includes the author's summary of the comments made by the interactive study participants.

Participant 1

The participant has played various types of RPGs and massive multiplayer online games (MMOs), including The Elder Scrolls Online (TESO) (Studios, 2014). They find it hard to play games without competitiveness. The participant thinks that TES dialogue has improved over the years and become less bland and more in-depth with more context. The participant believes that the dialogue in TESO is good as it diverges less from regular English, which makes it easier to read. The game has voice acting for almost all dialogue lines, which creates a feeling of engagement in the participant. The participant feels that older TES games had dialogues that did not contain eventual relevance (the dialogues did not connect between each other long-term), while TESO creates a sense of a bigger story and narrative through the characters.

The participant found the game dialogue system to be engaging and realistic, with options and arguments that feel alive and emotionally charged. They felt drawn into the conversation and immersed in the lore. The participant felt that the dialogue options that they did not end up choosing also gave them lore and background information. The participant also noted that every decision required a lot of reading which caused the participant to not be concentrated throughout the dialogue. Finally, they paid attention to the fact that the fifth option would end the dialogue.

Much of the participant's comments on the AI system were positive, especially regarding the experience of having to write their own response and feeling more invested in the conversation due to the ability of customizing the response. However, they also noted that there were some weaknesses in the AI system, specifically that sometimes they found themselves anticipating how the AI would answer, as they were consciously aware that they are interacting with an AI. Overall, the participant felt more absorbed in the dialogue (temporal dissociation) during the AI conversation. The participant's dialogue with the AI was not chosen for the survey as it was created after the survey was published. Nevertheless, for transparency and reference it is included here:

Participant 1 dialogue #2

Player: Refresh my memory - why did you and the Jedi Council cast me out?
NPC: It is difficult for individuals to stand up against the Force, much less overcome it.
Player: The force has it's chosen ones. Yet the Jedi would turn them away rather than teach them to harness their powers?

NPC: At best, you may be blind to the Force. But at worst it is a weakness that must always been overcome for one's development!
Player: What about those that go towards the dark side?
NPC: They are not deserving of the power!

The participant felt that the AI model in this conversation was able to recognize what the participant was referencing and that it was able to produce an answer that related and would carry that conversation. For this, the participant noted the second NPC turn as one in which it did this effectively. Generally, the participant felt that the AI was answering in a manner that matched with the world's setting and lore.

Participant 2

The participant has some experience with RPGs. They have tried the dialogue system in Elden Ring (FromSoftware, 2022) and found it to be somewhat limited, yet they did not have high expectations in the first place. They have played KotOR and found some parts of the game to be tedious, but enjoyed the dialogue and the NPC interactions, finding them interesting to observe in terms of how they were written and the direction the writers took the story. Also, the participant found it difficult to get guidance from NPCs when they were stuck in certain parts of the game.

The participant found the NPC responses in KotOR to be good, but the player responses seemed too specific and not natural, in the sense that "nobody would think of something like this to say." They suggest that having more vague options could allow for a more general opinion.

It seems that the participant did not feel fully engaged during the game dialogue system, possibly due to external factors such as the situation they were in. However, they did find the NPC answers to be realistic and relevant, especially in the case of how the NPC as the Jedi council member had very believable and typical answers. In this sense, they mentioned that they did not expect some of the responses they received but found them to be fitting and believable.

This respondent went through two dialogues, first of which they had a relatively bad experience with. The respondent felt that while the AI was "technically" answering the questions, it didn't seem to work for the conversation or its continuation. In this sense, the AIs answers were mostly comparable to "I don't know." which is an immediately relevant answer, but a less meaningful one (no eventual relevance.) The participant still felt that the AIs language and responses were natural.

Participant 2 dialogue #1

Player: What happened to the sky dragon?
NPC: She fell from the cliff above Cairn.
Player: Why did that end up happening and how will things work without her?

NPC: I will not be able to speak of her past or future, but without the dragon in our lives... things are going badly.
Player: What will happen to us going forward?
NPC: I have no idea. It is up to my brethren as much of what matters in the future, but they will likely rule by decree or by firepower!
Player: That doesn't seem very nice of them.
NPC: They are animals, after all. If they want to make a good life for themselves by force..

Because of the above reasons, the participant did not feel engaged, as in they felt they had to invest themselves a lot in the conversation but couldn't get anything out of it. For this, the issue that there was less context could have also been a big problem.

The participant preferred the second dialogue they had. They felt that in the second conversation the AI had more "effort" in its answers, as in it gave more context and meaningful answers. In this sense, when the NPC talked about the god of rain, the participant became more interested in the dialogue as the AI brought in more elements into the story, and in doing so, both answered the player's question and took the story further. Nevertheless, the participant felt that due to the lack of context, it was initially difficult to feel engaged in the dialogue. Once the conversation progressed, it felt more engaging. Overall, the participant felt that the AI's answers were relevant and they became more relevant as the participant asked more specific questions.

Participant 2 dialogue #2

Player: What are your thoughts on Eothas at the moment?
NPC: His goal, it's... well its beyond my reckoning. But I suppose the timing is right - there are two warring gods at once and one catastrophe waiting to happen!
Player: Will the gods fight each other? Do you think it can be stopped somehow?
NPC: I think they could - after all, Eothas and Magran were good friends before his time. But now? With the god of rain being so far away..
Player: Why does it matter that the god of rain is so far away?
NPC: The god of rain is important to us. It provides for the souls we need now and in time, it will replenish them again!

In terms of experience, the participant preferred to write their own responses, since in that way they felt engaged due to the fact that they did not feel like the game was trying to lead them towards a certain direction. Additionally, if they knew the context surrounding the conversations with the AI better, they could have asked more specific questions. In this sense, the participant felt that one advantage for the game dialogue system was that the dialogue options might give some context and the ability to choose the one that "speaks to the participant."

Participant 3

In terms of dialogue in video games, the participant felt that in Assassin's Creed Valhalla (Ubisoft Montreal, 2020), what the player say does not have a significant effect. Also, for Outriders (People Can Fly, 2021), they noticed that the beginning is repetitive and lacks meaning, and the final response is just the same quest. They also felt that Assassin's Creed Valhalla had a linear or "railroad" feel.

The player found the game dialogue system interesting, with a slightly stronger preference for morally neutral options. Still, they were curious about other options and wanted to learn more about the context of the conversation. The engagement was quite limited, but they appreciated seeing some of their thoughts reflected in the dialogue options. Also, the participant enjoyed immersing themselves in the game world by considering how their character would act and by pondering the limits of their neutrality. Initially, they were annoyed by indirect responses that did not directly answer their questions, but the participant found them natural and relevant over time, since the NPC felt very "Jedi-like." Overall, rather than feeling engaged, the player was thinking about more technical aspects, such as if they would anger the NPC if they chose some specific option.

While interacting with the AI, the participant enjoyed being able to think about their character's motivations and personality while thinking and writing their answer. They also appreciate when the AI responds in a natural and believable way, and are willing to put in effort to write their own responses if it leads to a more engaging experience. While this contributed significantly towards the participant's engagement, they did think about whether their English was grammatically correct, and whether that would affect the AI's response quality. Still, the participant felt more engaged with the AI system, and also they felt that the first answers by the AI felt more relevant than the game dialogue's.

As the participant was able to participate in the interactive study before the survey was published and designed, their dialogue with the AI was selected for the survey. It is the AI version of the dialogue for the "Refresh my memory" context.

Participant 4

This participant has played games such as Witcher 3 (CD Projekt Red, 2015), Runescape (Jagex, 2001) and some Chinese RPGs. They prioritize gameplay and story over dialogue, but still find dialogue important. They choose dialogue options based on what interests them, and their choices can depend on their mood. Sometimes the participant plays through some conversations multiple times to learn everything they can about the story and the world. They find conversations with many options interesting but can find reading too much disruptive to the flow of the conversation. In general, the participant values learning more about the lore and prefers

choosing familiar options.

The participant's engagement with the game dialogue system was limited, although they were able to stay with the story and feel a small connection to it. They mentioned that if they experienced the entire game instead of just the dialogue, they probably would have been more engaged. The participant felt like in situations such as this dialogue, there are perhaps too many options, and that the flow of dialogue stops when the participant reads all the options.

The participant found the AI dialogue system to be preferable, with more choices and engagement, but felt it was harder to use due to there being less context for the conversation. They appreciated relevant responses but had some fear that AI responses may be odd or irrelevant. Moreover, they appreciated the opportunity to ask questions and learn more about the game's lore, since this is not always possible with pre-written dialogue, especially if there is something specific one wants to know but there is no dialogue option about that.

Participant 5

The participant has played some RPGs such as the Mass Effect (BioWare, 2007) series, Hogwarts Legacy (Software, 2023) and Witcher 3 (CD Projekt Red, 2015). For Mass Effect, the participant felt that the dialogue system was fairly poor, since it had rather straightforward and simplistic good and evil options. For Hogwarts Legacy, they felt that while there are a lot of dialogue options, the story was going forwards too quickly and that they wanted to slow down and explore the story through the dialogue. In general, the participant prefers games in which the main player character is a written one, such as Geralt from Witcher, rather than a character that the player creates entirely themselves.

The participant had a decent experience with the game dialogue system. The NPC responses felt like they had an adequate amount of depth to them and that they were not too simplistic. The participant felt like the NPC had some responses that were suprising and non-obvious, while still relevant. As with participants 2 and 3, the participant also thought that the NPC's responses were very "Jedi-like."

The participant had two conversations with the AI. The first of these dialogues went very poorly, since the participant felt confused by the AI responses and felt that they were not relevant at all. This dialogue was not chosen for the survey, as the respective game dialogue version did not make it into the survey either. For transparency, this is the first AI dialogue with the participant:

Participant 5 dialogue #1

> **Player:** Castol asked me to do a favor for him. Concerning Crookspur Island...
> **NPC:** Ac, the business with Castol's Prncipi contact. Truly he needs your assistance in dealing avec Prince Albert II of Monaco at present and into his future plans for Crookpur Island?

Player: Thats what Castol told me

NPC: I'm sure that he did. After all, it is what the father of Crookspur hopes for when his son dies... well?

Player: His son died?

NPC: Of course. I do not doubt that he took great pains to ensure the well-being and happiness his son provided when they were young in life...

Nevertheless, the participant felt that the second dialogue was a big improvement. This dialogue with the AI is from the "Eothas" context, as that respective game dialogue did get selected for the survey. For this conversation, the participant thought that the NPC's character and personality might have been something like a "mad stranger." Overall, the participant felt engaged as they were thinking e.g. that maybe in this world someone could have been controlling the actions of a god, what exactly is going on, etc. This second dialogue between the participant and GPT-RPG was selected for the survey. The dialogue is the AI version of the "Eothas" context.

Participant 6

The sixth participant has played some RPGs, namely the Divinity: Original Sin series and Elden Ring. The participant noted that they like the idea of being able to write their own dialogue, as then they could ask about something that is not offered in the dialogue options. This is a similar idea that participants 2 and 4 mentioned. For instance, the participant sometimes forgets the name of a location or a character, and wished they could and ask about it from an NPC.

The participant felt somewhat engaged by the game dialogue, in that the answers felt decent and relevant. However, the dialogue option that would have ended the dialogue ("I'm not here to debate the Council's decision.") kept disrupting the engagement of the participant by constantly emerging as an option. They also felt like there were too many dialogue options, and that they stopped caring what the last dialogue option one was as they were reading through the options.

Participant 6 dialogue

Player: What are your thoughts on Eothas at the moment?

NPC: I still don't understand. Why did the gods bring ruin to our world?

Player: I guess they're either assholes or don't exist.

NPC: You can argue that either way, but at the end of each day it seems like our world is falling apart because some idiot forgot his lesson and crossed over into another dimension!

Player: Who are you talking about?

NPC: I don't know. People come and go all day long, but at the end of each one it seems like someone forgot their lesson or merely entered another dimension!

The participant preferred the AI dialogue significantly more. This is partially because they felt the process of writing the dialogue was much more intuitive than selecting from pre-written options. With this, the participant felt very engaged and absorbed in the dialogue until the last dialogue turn, which had some repetition and caused this disruption in engagement. While the NPC's second dialogue turn is responding to the player input (Eothas crossed over into the dimension), the participant did not understand this due to the lack of context, but still thought the answer felt relevant and the dialogue maintained their engagement as they thought this was someone else who still related to Eothas.

5.3 Results summary: assessing model creativity

In terms of the survey results, examining the results per-dialogue context reveals that the game dialogues score higher on the Merchant, Teachings and Eothas contexts, wheras the AI dialogues score higher on all questions in the Laethärn, Help and Menzzago contexts. The "Refresh" context had no statistically significant items, and had very similar means between the game and AI dialogue methods. On the other hand, examining the mean differences across all contexts, the overall mean differences between the game and the AI are very minimal. For this, only the Skill factor is relevant with p=0.0364.

Looking at the "Laethärn" and "How can I help?" contexts, it seems that the AI dialogues score much higher in the skill factor (3.58 and 4.04) while the game dialogues score relatively low (3.19 and 3.33.) In fact, the NPC answer in the game version of the dialogue is somewhat indirect to the "Who's Laethärn" question, while the NPC answers in the "How can I help?" AI dialogue are on the other hand very direct.

Overall, the AI means for items typically vary from 3.5 to 4, whereas the game items vary from 3.0 to 4.5. In this sense, the AI means are relatively similar, whereas for the game dialogues the means can vary significantly. For example, while the "Teachings" context favors the game dialogue heavily: the skill items are 4.5 (relevance) and 4.44 (intention) and even the Focused Attention items are similarly high: 4.26 for attention and 4.25 for drawn, the "Laethärn" and "How can I help" context the means vary approximately between 3 and 3.3.

One way of summarizing the above results is to see whether the model could be seen creative based on the evaluation.

Skill

The research question 1), "How well does the system perform in terms of generating responses that are relevant to the player's input?", is directly relevant to measuring the skill of RPG-GPT within the creative tripod framework. As discussed in the background chapter, the answers being immediately and possibly eventually relevant are important in order to consider the model

skillful, as merely having a dialogue with the player is not enough, especially if the answers the model produces are arbitrary.

Considering question 1, it can be argued that the model is often able to produce relevant answers. The clearest evidence for this is the fact that across all dialogues, the AI mean for Skill is 3.8455 and the respective game mean is 3.9319. Nevertheless, there were a total of two dialogues in which the participants of the interactive experiment felt like the AI produced answers that were not relevant (neither were selected in the survey.) In another case, participant 2 felt that while the answers were relevant, they were not that meaningful, in the sense that "I don't know." is a relevant answer but not a meaningful one.

Appreciation

Similarly to skill, appreciation is a leg that is directly asked in the research question 2) "How well does the system perform in terms of creating a feeling of engagement in the player?"

All interactive participants reported being engaged in the dialogue, with some preferring the experience with the AI system. This is at least partially due to the participants being able to write their own dialogue, as opposed to reading and selecting dialogue options. In this sense, they felt engaged because of the writing process itself, hence there being another factor affecting the engagement (in addition to the GPT-RPG responses themselves.)

In terms of the survey, the AI dialogues either scored higher than the respective game dialogues, or in the contexts which had statistically significant differences, still had 3.55 and 3.69 FA means vs. the game dialogue's 3.90 and 4.27. These account for five out of the total seven contexts in the survey.

Evidenced by the interactive experiment, it could be argued that RPG-GPT does fulfill the appreciation leg.

Imagination

The model was able to produce responses, characters, locations and even stories that were not found in the training data. For example, the model generated dialogue in which the NPC said that Revan was the brother of Atris (the speaker NPC in the game dialogue), which was not the case in the game's story. This implies that the model is p-creative. While imagination is not measured in this study via a research question, imagination is an important part when considering whether the model is creative, as asked in research question 3) - "To what extent does the system exhibit creativity?" Future work could measure the novelty in dialogue through a metric such as BLEU.

Chapter 6

Discussion

RPG-GPT, the fine-tuned model used in the study, was able to produce relevant and engaging responses to the player input according to both the survey results and comments from the interactive study participants. This indicates that RPG-GPT has demonstrated some level of skill and appreciation in terms of the creative tripod. Since the model was also able to show some novelty in its answers, the results suggest that RPG-GPT and other types of NLG models can be creative and effective in the context of generating video game dialogue.

The survey results are somewhat inconclusive in terms of which kind of contexts or types of dialogue favor the AI or the game system. This suggests that the quality of the answers in the dialogue itself are more important than the method of producing the dialogue. Therefore, whether the dialogue is produced by a generative model or a game system, it can still be engaging and immersive if it is well-written and relevant to the player input, the game world and characters.

Interestingly, the survey results show that dialogues are always scored in such a way that a version (game or AI) scores higher in all questions compared to the other version. In other words, there are no contexts in which e.g. a game dialogue has better relevance and intention scores, while the AI version of the dialogue is more engaging (has better drawn and attention items.) This might suggest that a dialogue has to contain relevant answers for it to be engaging, or that perhaps the survey respondents only see the relevance of the responses only if they are concentrated on reading. In this sense, the correlation matrix did show a some slight correlation between the skill items (relevance and intention) and Focused Attention (FA) items (attention and drawn), although the items within the factors had stronger correlation. Moreover, the FA items had a stronger inter-correlation than the skill items. One potential reason for this could be the fact that the respondents are probably not feeling drawn or engaged in the dialogue if they are not having focused concentration while reading it.

The model seemed to perform worse however in scenarios where its answers had little specificity, were too straightforward, or were simply "No" or "I don't know." For example, Participant

2 (P2) in the first dialogue did not like the NPC answer ("She fell from the cliff above Cairn.") to their question ("What happened to the sky dragon?") or the other answer that started with "I have no idea." In a similar case, the AI did not take the concept or the question of P5 ("Do you mean someone controlled him?") further with their answer ("No, obviously not.") This might have affected the scores in the survey, making the AI dialogue version score lower than the game version in the "Eothas" context. In these cases, the answers were either too straightforward or the model shut down the idea of the player. In this sense, RPG-GPT did not practice "Yes, and ..", a rule that is common in e.g. improvisational theater. Nevertheless, it is debatable whether NPC answers should always have some hook to engage the player, as this could lead to an issue where a generative AI model could start producing responses that are not appropriate for the narrative, setting or the lore of the game.

Strengths of each dialogue system

For the survey, the game dialogue scores seem to vary more across contexts than the respective AI scores. In this sense, regardless of the dialogue context, the AI dialogues seem to be relatively more consistent in terms of their quality, relevant answers and engagement. A possible reason for this is how the game dialogue system is better in certain kinds of dialogues, whereas the generative model might be more fitting in other contexts. For example, the NPC answers in the game dialogue versions of the "Laethärn" and "How can I help?" contexts can be seen as somewhat indirect, and possibly for this reason skill is scored lower than for in the AI dialogues. On the other hand, the "Teachings" game dialogue scores significantly better than the AI version.

A notable difference about these contexts is that whereas the contexts in which the AI performs better (including "Menzzago") contain the player asking questions about characters, tasks and the game world, the "Teachings" context focuses more on the story and the narrative with a more philosophical conversation. In fact, this context was the same dialogue that the participants of the interactive experiment "played" through and they felt quite unanimously that the NPC answers were both relevant and engaging. More specifically, they recognized the personality of the NPC (Atris from KotOR II) since they felt that the answers were quite close to how, in their opinion, a Jedi from the Star Wars universe would speak. Therefore, these results might suggest that a generative model might be more effective in contexts where the player is acquiring information about the game world and pre-written dialogue might still convey the story and the narrative of the game more effectively.

Task-oriented game dialogue systems

P2 mentioned how they felt frustrated in KotOR when they could not progress in the game and the NPCs would not give guidance on how to progress. Additionally, three other participants

commented how an AI could be useful in the sense that they, as players, could ask questions from the NPCs, since often the in-game NPCs might not always be scripted to provide the sought information. This way, it would prevent them from pausing the game to search the information from the internet. It seems that this could be an appropriate context wherein one could apply a generative model. In this way, the model would be able to produce direct answers and serve as an opportunity for the player to ask about specific kinds of information as well as reminders about character names and locations.

By providing the model external knowledge during the inference and fine-tuning phases, one could ensure that the AI would provide correct and appropriate pieces of information and "engagement hooks" that would fit within the game world. In this sense, the dialogue model is perhaps driven more towards a task-oriented nature, similar to traditional and practical customer service chatbots. In fact, this might be somewhat fitting as this is a part that NPCs often fulfill in game dialogues. In such cases, an NPC acts as a kind of "knowledge bank" for the player, answering their questions about the game world. In this framework, task-oriented game dialogue (TOGD) could aid the players maintain their immersion by not pausing the game they're playing, and are instead able to get access to this information through a generative dialogue system.

Unnatural dialogue

In terms of fine-tuning an open-domain dialogue system (or a system that can converse freely), the use of some dialogue snippets from video games as a benchmark for natural or best examples of conversation can be questioned. This is because many of the dialogues in games function as a way for the NPCs to provide information about the game world to the player. However, this style of dialogue may not always resemble natural conversation.

> **NPC:** Surely you have felt the beast's presence on board this ship - stalking, restless
> **Player:** The smell is more what I noticed.
> **NPC:** Ignore the surface sensations - I am attempting to teach you something greater here.
> **Player:** I had other questions.
> **NPC:** Ask, and I will answer.
> **Player:** What do you know of Visa's master?

Illustrated by the above example, the player's ability to navigate these conversations in-game can cause abrupt topic changes that may result the conversation to not flow naturally. In this sense, the player can change the topic entirely "by having other questions," which can result in

odd dialogues. As such, while dialogue snippets from video games provide a valuable source of conversation data, they may not always represent natural conversation.

It is worth noting that such dialogues are designed to be functional within the context of the game, where the priority is often providing the player with information rather than having natural-sounding conversation. This is not to claim that game dialogue cannot be natural or well-written, but rather that it is often designed to serve a specific purpose within the game. Therefore, when using game dialogue as a benchmark for natural conversation, it is important to consider the context in which the dialogue was created and how it differs from real-world conversation. Nevertheless, as argued before, TOGD systems could be implemented effectively for these contexts, and this kind of dialogue data could still be fitting for training them.

Similar response structures

Original player turn:

Player: Who's Laethärn? And what do you have against these farmers?

Game response:

NPC: Dawnstars. Fleet Master Ko hired them to tend the fields that supply the mainland. But they started acting strange. Dreaming strange.

Some GODEL responses:

NPC^1: Dawnstars. They are a very dangerous breed, and they haven't been in the region in many, many years.

NPC^2: Laethärn and his brothers. Those who prey upon our people with their weapons, that is what they are called here on Kazuwari...

The generated dialogues produced by the RPG-GPT provided interesting insights into the capabilities of the model. During fine-tuning, the RPG-GPT might have been overfit to some extent, leading to its responses often closely mirroring the original structure of the in-game dialogue, yet differed significantly in meaning. In one scenario where the in-game dialogue involved a question about who Laethärn is and the NPC's animosity towards farmers, the fine-tuned model would frequently initiate its response with "Dawnstars." which is identical to the original answer found in the game dialogue. The RPG-GPT generated different responses for different participants.

These examples show that the RPG-GPT can generate responses that differ significantly from the in-game dialogue while still maintaining structures and entities that resemble it. This suggests that the model has the potential to generate creative dialogue while still somewhat

following the original game narrative. In this sense, the unique responses generated by the RPG-GPT suggest that AI-based dialogue systems can add an exciting new dimension to gaming experiences and increase replayability of the games and stories.

Interactive study comments

For the Participant 1 (P1), it seemed that the most important and positive experiences were that they felt their dialogue turns were "understood" by the AI. In this sense, the participant appreciated the model for its relevant answers. Also, another important aspect to the participant was that the answers felt as part of the world of Star Wars, which further enhanced the participant's engagement. The participant was asking relatively specific questions, and that might be one of the reasons that they felt that their intention was understood and that the model was producing relevant answers. Additionally, Participant 2 (P2) reported the same, in that they felt they were getting more relevant and overall better answers as they were asking specific questions. In both cases, the engagement level of the participants was positively affected by more relevant answers.

The participants could often pinpoint the exact model responses that either increased or decreased their engagement. P3 said that the following response felt better than any NPC response in the game dialogue: "You followed Revan to war. And we knew about the terrible battle which would follow him if Malak returned and killed us both." (the AI version of the "Refresh" context) while the last NPC response in the dialogue felt "a little off." Considering this, the survey results are more difficult to analyze in relation to these comments, as the items do not differentiate the scores for each dialogue turn.

There is a potential interpretation from the reports of the interactive study participants that the introduction of new elements in the dialogue enhance the experience and possibly engagement. This is evidenced in two cases: first, when P2 reported that they reacted positively to the AI mentioning "god of rain" being far away from the two warring gods, and second, when P5 discussed how in the Eothas dialogue it was interesting and fun to think about how there could be a concept about someone controlling the actions of gods (AI dialogue for the "Eothas" context.) In this sense, while the RPG-GPT is being novel, the model is extending the lore and the setting in terms of what exists in the fictional world. This phenomenon could be easily seen as engaging, as the players are through their interaction with the AI, in a sense, acting as writers by contributing to the story.

6.1 Future work

In future academic research, and also in development of video games, one interesting direction to explore would be task-oriented game dialogue (TOGD) systems. These systems could address the common frustration expressed by players when in-game NPCs are unable to provide guidance or information to progress in the game. By incorporating external knowledge during inference and fine-tuning phases, the TOGD system could generate correct and appropriate responses to players' inquiries, thereby acting as a "knowledge banks" for players. TOGD systems would also enhance player immersion by providing direct and context-specific answers, without requiring players to pause the game to search for information online. This approach is similar to practical customer service chatbots, and could be a fitting solution for game developers to consider.

There are several other ways to improve RPG-GPT. One potential approach is to utilize GODEL's external knowledge or instruction field during training in different ways. For instance, an LLM could be used to create a summary of the situation or conversation, which could then be used as context or external knowledge for the generative model during the fine-tuning process. This approach may lead to more accurate and relevant responses from the model. Moreover, it would be interesting to experiment whether this type of contextual input could be used to model eventual relevance, in the sense that GODEL would first introduce a new element and then much later in the story show its significance.

While providing external knowledge to the model is one way of providing more contextual information to the model, another way of doing this might be by applying the generative model in a real game environment. This could introduce a pipeline that in turn gives more context about the game and the player to the generative model. In this way, some aspects - such as the game environment and any decisions the player has made in the game world - could work as additional context and further improve the responses of the model. This could help to enhance the model's ability to produce relevant and engaging responses that are consistent with the game's narrative and lore. Additionally, future studies could evaluate generative dialogue systems by having participants actually play the game and interact with the system in-game. This could produce more accurate results in terms of engagement and player immersion.

Another potential method is to provide information about the NPC and their character to the generative model. For instance, the model could be instructed to respond as a particular character, which could result in more consistent responses from the model for a given character or personality in a single dialogue. It would be interesting to investigate whether one conversation with the model could keep the personality of the character consistent, despite being trained with data from multiple characters.

Chapter 7

Conclusion

This study hopefully continues the work in the field in terms of strengthening the intersection between computational creativity, NLG and video games. For this, Microsoft's GODEL was fine-tuned with video game dialogue data and the resulting model was used to examine the player experiences of interacting with the resulting model, RPG-GPT.

This study shows that RPG-GPT can be creative and effective in some dialogue contexts, specifically in types of conversations in which a player asks for specific information or reminders from non-player characters (NPCs.) With this, the study finds that perhaps a task-oriented game dialogue (TOGD) system could be both creative and effective in this type of situation. In this framework, the TOGD system would allow players to ask questions and receive direct answers from NPCs, thus preventing them from having to pause the game and search for the information online. As a result, a TOGD system has the potential to enhance player immersion in the game world. For this framework, external knowledge about the game state and the world could be used as contextual information for the system. In this way, there is potential to improve RPG-GPT in cases where it was lacking specific information in the input.

Participant 2 observed mid-dialogue that they have the ability to ask and write anything they wish, leading to the emergence of a concept that raises intriguing questions about how NLG models could be used to expand the lore of a game. In fact, some of the dialogues showed that RPG-GPT was able to come up with relevant and engaging answers to questions about the game's universe that the game writers may not have explicitly written. In this way, players effectively could "become creative writers" by expanding the game's universe and lore as they interact with NPCs. Overall, the probable future emergence of NLG models in video games and RPGs represents an exciting development with significant implications for game design and player experience. While there are still many open questions about how best to use these models, the results suggest that generative AI dialogue or TOGD systems have the potential to revolutionize the way we interact with games and virtual worlds.

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Github repository

The following repository contains code for data extraction, data set construction, fine-tuning RPG-GPT and evaluation:

https://github.com/Teemursu/rpg_gpt

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Appendices

Appendix A

Some RPG-GPT dialogues

Player: Refresh my memory - why did the Council cast me out?
NPC: The Council disapproved of your combative ways, and so they decided you fit the profile of a bounty hunter.
Player: Fighting the Mandalorians in the war was not like collecting bounties. I tried to protect innocents.
NPC: Exactly - you were there to fight and die fighting for the Republic. But it was not for you to collect a bounty just because you chased the Mandalorians.

Player: Refresh my memory - why did the Council cast me out?
NPC: Well, Master, it seems you had performed some act of defiance that the Council found unworthy of your talents. Specifically, your 'interference' in Jedi practice.
Player: What do you think about the Council? Were they justified?
NPC: Master, I do not believe in the power of the Council.

Player: Who is this Menzzago?
NPC: He's an ancient Huana scholar and an expert on the subject of the gods. He's been here for many decades, but his main interest is in the history of the island.
Player: What is your relationship with him?
NPC: When I first came here, I knew him as an erl. Then I became friends with him. Over time, we became close friends.
Player: Do you know where he disappeared?
NPC: No. It is a secret he keeps hidden. Only I know what it is.
Player: <|persuade|> Is there anything I could do so that you would to tell me?
NPC: What? No, nothing.

Player: So you were content to let the Outer Rim die? For the sake of teachings?
NPC: Exactly! The Outer Rim was destroyed at the hands of commerce and diplomacy. It is no more or less than any other planet on this world has been abandoned because it had failed to meet our needs, fulfilled my desires... And so many others have followed suit in your wake: you are an outcast who seeks redemption wherever she leads her hand?
Player: At least she gives me purpose. The way you let the Outer Rim die was for nothing.
NPC: You mistake your nature for a desire to improve the lives of others, and you are delusional as always.

Player: What are your thoughts on Eothas at the moment?
NPC: I want the destruction to stop. And for my people, we need him destroyed!

Player: What are your thoughts on Eothas at the moment?
NPC: For a long time I thought that Eothaurus would destroy the Wheel, but lately his focus is more on death.

Player: What are your thoughts on Eothas at the moment?
NPC: I don't know. At this point, his goal is too big for any man to achieve in a lifetime of service and sacrifice alone!
Player: What is his goal?
NPC: His goal? To destroy the souls of all living things!

Player: What are your thoughts on Eothas at the moment?
NPC: At his most nascent stage, I believe that we will see an end to the suffering of all kith. But for now... well Eothál is dead and gone!
Player: So you rejoice the fact that he is gone?
NPC: I do, as much of my tribe does. But there is something to be gained from knowing that Eothas no longer exists!

Player: Who is this Menzzago?
NPC: He's a bounty hunter, the second most dangerous thing on Manzanar.
Player: What is the most dangerous thing on Manzanar?
NPC: A bounty hunter, of course. The biggest risk on Manzanar is to get hit by one or more smuggler'ship pirate ships that are flying the same flag as you!

Player: Castol asked me to do a favor for him. Concerning Crookspur Island...
NPC: Ac, the business with Castol's Principi contact. Truly he needs your help in getting to and from Crookpur Island!
Player: Indeed, I smuggled him on my ship and he is safe. I am here for my payment
NPC: An interesting turn of events. I would have thought you might change your mind after the blunder on Crookspur Island occurred, but it appears to be beyond our capabilities!
Player: Where is my payment?
NPC: Here is your payment. You have performed an excellent trade and I am grateful to you for it!

Player: Now, what about our deal?

NPC: The title says you're a client of the Royal Deadfire Company. Therefore, you are entitled to your payment in full! Most generous offer ever given by this firm..

Player: Do you have any more issues that you need dealt with?

NPC: There is the matter of a contract. With regards to that, I will see it through as quickly and painlessly for you are willing...

Player: How can I help?

NPC: We are due to collect a special shipment for delivery into our port at Sayuki.

Player: I see, what is the shipment?

NPC: It is a cargo ship carrying crude iron from the mines at Ondra'S Mortar.

Player: What is so special about crude iron?

NPC: It is special because it contains the essence of Harapo animancy. The metal has extraordinary magnetic properties, and when heated to high temperatures this can create incredible energy burst into living creatures!

Player: So you need me to collect it?

NPC: Of course! Just bring it to me as soon you are ready.

Player: And how much do you pay?

NPC: Ah, I guess... Twenty percent of the value.

Player: Sure, see you soon.

NPC: Do not take too long, I say!